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Join-semidistributive lattices and convex geometries

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Abstract

We introduce the notion of a convex geometry extending the notion of a finite closure system with the anti-exchange property known in combinatorics. This notion becomes essential for the different embedding results in the class of join-semidistributive lattices. In particular, we prove that every finite join-semidistributive lattice can be embedded into a lattice $S_p(A)$ of algebraic subsets of a suitable algebraic lattice A . This latter construction, $S_p(A)$, is a key example of a convex geometry that plays an analogous role in hierarchy of join-semidistributive lattices as a lattice of equivalence relations does in the class of modular lattices. We give numerous examples of convex geometries that emerge in different branches of mathematics from geometry to graph theory. We also discuss the introduced notion of a strong convex geometry that might promise the development of rich structural theory of convex geometries.

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A lattice $(L, \vee, \wedge)$ is called *join-semidistributive*, if

$$x \vee y = x \vee z \text{ implies that } x \vee y = x \vee (y \wedge z) \text{ for all } x, y, z \in L.$$

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Join-semidistributive lattices were introduced in 1961 by Jónsson [46] in his studies of free lattices. Nowadays, the class $\mathbf{SD}_\vee$ of join-semidistributive lattices, together with the class $\mathbf{M}$ of modular lattices, are among the most important extensions of $\mathbf{D}$, the class of distributive lattices. (Observe that $\mathbf{SD}_\vee \cap \mathbf{M} = \mathbf{D}$.) The investigation of join-semidistributive lattices originates in the study of objects such as free and lower bounded lattices as well as their sublattices [27] or lattices of quasivarieties [31,33]. Join-semidistributive lattices also have many applications in various branches of lattice theory and universal algebra such as the theory of varieties of lattices [44,48,57], tame congruence theory [28,42], and the theory of lattices with irredundant decompositions [17,32].

It is known [37] that a key to the study of modular lattices is given by *combinatorial geometries*, that is, modular geometric lattices and projective spaces. One of our main goals is to show that the latter geometrical objects have a counterpart in the theory of join-semidistributive lattices. This counterpart has already been christened *convex geometries* in combinatorics a long time ago, and the antithesis with matroids clearly recognized. Namely, the role played by the *exchange property* for matroids is played by the *anti-exchange property* for convex geometries. More precisely, while matroids are lattices of closed sets of closure spaces with the exchange property, *antimatroids* are lattices of open sets of closure spaces with the anti-exchange property, see [15]. In the present paper, a *convex geometry* is a closure space with the anti-exchange property, see Definition 1.6. To our knowledge, the lattice-theoretical facet of convex geometries appears in full only in the present paper, especially as far as *infinite* convex geometries are concerned. This will justify further the often encountered juxtaposition of join-semidistributivity and modularity.

The first source of finite convex geometries was found in lattice theory. In 1940, Dilworth [17] discovered these objects as a special class of those finite (lower) semimodular lattices that have unique irredundant decompositions. Later on, finite convex geometries were rediscovered many times as the abstract framework underlying the principal features of some combinatorial constructions (see a review of this topic in [56]). The systematical study of finite convex geometry as an abstraction of the notion of convexity in Euclidean space was launched by Jamison [43] and Edelman [20], and as the special greedoids by Korte and Lovász [49]. For an update state of this theory, the reader should consult [21,50].

According to one of the characterizations of finite convex geometries (see Theorem 1.9), a finite lattice is the closure lattice of some finite convex geometry iff it is join-semidistributive and lower semimodular. In Section 1, we shall prove (Theorem 1.11) that any finite join-semidistributive lattice can be embedded into some finite atomistic join-semidistributive lattice, or, equivalently, into the closure lattice of some finite *atomistic* convex geometry. Together with Theorem 1.4, this proves that the quasivariety $\mathbf{SD}_\vee$ of join-semidistributive lattices is generated by the closure lattices of finite convex geometries.

Another source of “universal” join-semidistributive lattices is the class of lattices of quasivarieties of general algebraic systems, the latter as considered in [52] or [34]. Unlike algebras, these algebraic systems may contain *relation symbols* as well.

Due to a theorem by Gorbunov and Tumanov [36], any quasivariety lattice is embedded into the lattice $S_p(A)$ of algebraic subsets of some algebraic lattice A . The lattice $S_p(A)$ is, in turn, isomorphic to some quasivariety lattice. It is worth noting that $S_p(A)$ is represented as a subquasivariety lattice of the class of pure relational systems, see [35].

In Section 2, we prove (Theorem 2.4) that an analogous embedding result holds for arbitrary finitely presented join-semidistributive lattices. This, together with results proved in Section 1, implies that the quasivariety $SD_\vee$ is generated by all quasivariety lattices. Observe that the statement obtained by replacing general quasivariety lattices by lattices of quasivarieties of algebras does not hold, see [28,33].

In Section 3, we discuss the various possibilities of defining a *strong convex geometry* in the infinite case. We first consider various often encountered classes of convex geometries. It turns out that all these examples are *atomistic*, and that they have either an algebraic or a dually algebraic closure lattice. Further, we compare the previously proved embedding theorems with the analogous results for combinatorial geometries. Among other results, we prove (Theorem 3.26) that any join-semidistributive lattice can be embedded into an atomistic, algebraic, biatomic convex geometry.

Despite our initial intention to complete the current paper before the second author's monograph [34] went into press, the order of the events has been reversed. As a consequence, the proofs of core results in Sections 1 and 2 in the present paper transcend their presentation given in the book.

Theorems 1.11 and 2.4 were also announced without proofs in [65].

We use in this paper the same notation and terminology as in [37].

1. Join-semidistributive lattices and finite convex geometries

1.1. Local theorem for the class of join-semidistributive lattices

We will several times require the following relativized version of join-semidistributivity.

Definition 1.1. Let L be a lattice, let A be a subset of L . We say that L satisfies $SD_\vee(A)$, if $x \vee y = x \vee z$ implies that $x \vee y = x \vee (y \wedge z)$ for all $x \in L$ and all $y, z \in A$.

In particular, L is join-semidistributive iff L satisfies $SD_\vee(L)$. We denote by $A^\vee$ the set of finite (nonempty) joins of elements of A . The following Lemma was suggested by F. Wehrung as a generalization of our initial version of this statement.

Lemma 1.2. Let L be a lattice, let A be a subset of L . Then the following are equivalent:

- (i) L satisfies $SD_\vee(A)$;

(ii) the following holds:

$$\begin{aligned} x \vee \bigvee_{i < m} y_i &= x \vee \bigvee_{j < n} z_j \text{ implies that } x \vee \bigvee_{i < m} y_i \\ &= x \vee \bigvee_{i < m, j < n} (y_i \wedge z_j) \end{aligned}$$

for all $x \in L$, $m, n > 0$, and $y_0, \dots, y_{m-1}, z_0, \dots, z_{n-1} \in A$.

(iii) L satisfies $\mathbf{SD}_\vee(A^\vee)$.

Proof. Implication (i) $\Rightarrow$ (ii) proceeds exactly the same way as the proof of [27, Theorem 1.21].

Implications (ii) $\Rightarrow$ (iii) and (iii) $\Rightarrow$ (i) are trivial. $\square$

We say that a lattice L is *additively generated by a subset A* if any element of L is a finite join of elements from A , that is, $A^\vee = L$.

Corollary 1.3. *Let L be a lattice additively generated by a subset A . Then L is join-semidistributive iff L satisfies $\mathbf{SD}_\vee(A)$.*

Now we are ready to prove the following embedding theorem.

Theorem 1.4. *Any join-semidistributive lattice can be embedded into an ultraproduct of finite join-semidistributive lattices. In particular, the quasivariety $\mathbf{SD}_\vee$ is generated by its finite members.*

Proof. Let L be an arbitrary join-semidistributive lattice and let M be a finite partial sublattice of L . Consider the meet-subsemilattice $M_\wedge$ of L generated by M and the join-subsemilattice $M_{\wedge\vee}$ of L generated by $M_\wedge$. We shall prove that $M_{\wedge\vee}$ is a finite join-semidistributive lattice with respect to the induced order, and that it contains M as a partial sublattice. Evidently, $|M_\wedge| \leq 2^{|M|}$, and the meet 0_M of all elements of M is the least element of $M_\wedge$. Analogously, $|M_{\wedge\vee}| \leq 2^{|M_\wedge|} \leq 2^{2^{|M|}}$. As 0_M is the least element in $M_{\wedge\vee}$, the latter is a lattice in which the joins coincide with the joins in L and the meets of elements from $M_\wedge$ are the same as in L . It follows that M is embedded into $M_{\wedge\vee}$ and $M_{\wedge\vee}$ is additively generated by its subset $M_\wedge$.

To prove that $M_{\wedge\vee} \in \mathbf{SD}_\vee$, we apply Corollary 1.3. If $a \vee x = a \vee y$ for some $a \in M_{\wedge\vee}$ and $x, y \in M_\wedge$, then, by join-semidistributivity, $a \vee x = a \vee (x \wedge y)$. As $x, y \in M_\wedge$, we have $x \wedge y = x \wedge_M y$. Therefore, $a \vee x = a \vee (x \wedge_M y)$.

Thus, any lattice $L \in \mathbf{SD}_\vee$ is locally embedded in the class $\mathbf{SD}_\vee^f$ of finite join-semidistributive lattices. By the local theorem due to Mal'cev, see [52, p. 164], L is embedded into some ultraproduct of systems from $\mathbf{SD}_\vee^f$. $\square$

By a *finitely presented join-semidistributive lattice*, we always mean a lattice defined within the class $\mathbf{SD}_\vee$ by a finite set of generators X and a finite set of relations in the variables X . Similarly, residual finiteness will be defined within the class $\mathbf{SD}_\vee$.

Corollary 1.5. *Any finitely presented join-semidistributive lattice is residually finite. Hence, the universal theory of $\mathbf{SD}_\vee$ is decidable.*

The latter statement means that there is a recursive procedure that decides whether a given universal formula of lattice theory is valid in every join-semidistributive lattice. Equivalently, there is a recursive procedure that decides, for a given presentation

$$P_1(\vec{a}) = Q_1(\vec{a}), \dots, P_n(\vec{a}) = Q_n(\vec{a}) \quad (1.1)$$

of a finitely presented join-semidistributive lattice L (where the P_i 's and the Q_i 's are lattice polynomials), whether an equation $P(\vec{a}) = Q(\vec{a})$ holds in L . Hence, the word problem for the join-semidistributive lattice with presentation (1.1) is solvable.

Proof. The first assertion follows from Theorem 3 in [12]. The second one follows from a general argument that probably first appears in [53], see also [52]. $\square$

Observe that in contrast with the results proved above for $\mathbf{SD}_\vee$, the variety of modular lattices is not generated by its finite members [25]. Moreover, the word problem is undecidable in free modular lattices $F_M(n)$ with $n \geq 4$, see [26,39].

Problem 1. For a finite partial lattice P , is the word problem in $F_{\mathbf{SD}_\vee}(P)$, the free join-semidistributive lattice on P , decidable in polynomial time?

By Corollary 1.5, the word problem in $F_{\mathbf{SD}_\vee}(P)$ is decidable. Moreover, it is well known that the word problem in finitely presented lattices, that is, in $F_L(P)$ for a finite partial lattice P , is decidable in polynomial time, see, for example, [27].

1.2. Finite convex geometries

In this section, we shall define convex geometries in general and discuss their basic properties in the finite case.

For any set X , we denote by $B(X)$ the set of all subsets of X . We recall that a map $\Phi: B(X) \rightarrow B(X)$ is a *closure operator* on X if for all $A, B \in X$:

- (1) $A \subseteq \Phi(A)$,
- (2) $A \subseteq B$ implies that $\Phi(A) \subseteq \Phi(B)$,
- (3) $\Phi(A) = \Phi(\Phi(A))$.

A pair (X, Φ) , where Φ is a closure operator on X , is said to be a *closure space*. We say that a closure space (X, Φ) is

- a *zero-closure space*, if $\Phi(\emptyset) = \emptyset$,
- *atomistic*, if $\Phi(\{x\}) = \{x\}$ for all $x \in X$ and $\Phi(\emptyset) = \emptyset$ (the latter condition is redundant unless X is a singleton).
- *algebraic*, if the equality $\Phi(A) = \bigvee \{\Phi(U) : U \subseteq A \text{ finite}\}$ holds for all $A \subseteq X$.

It is obvious that the set $\text{Cl}(X, \Phi)$ of all closed subsets of a closure space (X, Φ) , endowed with containment, forms a complete meet-subsemilattice of $B(X)$, that will be called the *closure lattice* of (X, Φ) . Conversely, given a complete meet-subsemilattice L of $B(X)$, we can define a closure operator Φ on X by the rule $\Phi(A) = \bigcap \{Y \in L : A \subseteq Y\}$, for all $A \in B(X)$. We say then that the closure space (X, Φ) on X is induced by the system L of its closed sets. Notice that we always can assume that Φ separates points in X , i.e., $\Phi(\{x\}) = \Phi(\{y\})$ implies $x = y$.

We observe that in the atomistic case, the closure space can be recovered from the lattice, as the points of X correspond to the atoms of $\text{Cl}(X, \Phi)$. Hence, we shall often identify an atomistic closure space with its closure lattice.

Definition 1.6. A zero-closure space (X, Φ) satisfies the *anti-exchange property* if the following statement holds:

$$x \in \Phi(A \cup \{y\}) \text{ and } x \notin A \text{ imply that } y \notin \Phi(A \cup \{x\})$$

$$\text{for all } x \neq y \text{ in } X \text{ and all closed } A \subseteq X. \quad (\text{AEP})$$

We then say that (X, Φ) is a *convex geometry*.

A standard example of a finite convex geometry is the following. Let X be a finite set of points in the n -dimensional Euclidean space $\mathbb{E}^n$. For an arbitrary subset $A \subseteq X$, we define $\Phi(A)$ to be the intersection of X with the convex hull of A in $\mathbb{E}^n$. Then (X, Φ) is a finite convex geometry and its closure lattice is denoted by $\text{Co}(\mathbb{E}^n, X)$.

A decomposition $y = \bigvee \{y_i : i \leq n\}$ of an element $y \in L$ into a join of join-irreducible elements $y_i \in L$ is called *irredundant* if $y > \bigvee \{y_i : i \leq n, i \neq j\}$ for all $j \leq n$.

Finally, a finite lattice is said to be *locally distributive* if for any $x \in L$ the interval $[\mathbf{x}', x]$ with $\mathbf{x}' = \bigwedge \{y : y < x\}$ is a distributive lattice, where $y < x$ means that x covers y .

We now recollect a few known characterizations of finite convex geometries.

The equivalence of statements (2)–(4) in the theorem below belongs to Edelman and Jamison [21] and the equivalence of (1) and (2) was proved by Dilworth [17]. The reference to other descriptions can be found in Monjardet [55,56].

Theorem 1.7. Let L be a finite lattice. Then the following are equivalent:

- (1) any element of L has a unique irredundant decomposition;
- (2) L is a locally distributive lattice;
- (3) L is the closure lattice $\text{Cl}(X, \Phi)$ of a closure space (X, Φ) with the anti-exchange property;
- (4) L is the closure lattice of a closure space (X, Φ) with the property that for any closed subset $A \neq X$ of X there exists $x \in X \setminus A$ such that $A \cup \{x\}$ is closed.

Property of a closure space (X, Φ) in (4) above is often accepted as the definition of a convex geometry in combinatorics.

We shall use later, at several places, the following well-known lemma.

Lemma 1.8. *Let L be a finite join-semidistributive lattice. For all $a, b \in L$ such that $a < b$, there exists a least element p of L such that $p \leq b$ and $p \not\leq a$, and p is join-irreducible.*

Proof. Put $X = \{x \in L : x \leq b \text{ and } x \not\leq a\}$; so X is nonempty. Put $p = \bigwedge X$. Then $a \vee x = b$ for all $x \in X$, whence, by join-semidistributivity, $a \vee p = b$, so p is the least element of X . In particular, $p > 0$. If $p = x \vee y$ with $x, y < p$, then $x \vee a = y \vee a = a$, thus $p \leq a$, a contradiction. Hence, p is join-irreducible. $\square$

The following description of finite convex geometries seems to have appeared first in explicit form in [19]. At least a part of the statement can be deduced from Theorem 1.7, but for the sake of completeness we give here the self-contained proof of this result.

Theorem 1.9 (Folklore). *A finite lattice L is isomorphic to the closure lattice of some finite convex geometry iff L is join-semidistributive and lower semimodular.*

Proof. Let (X, Φ) be a finite convex geometry. Suppose that for some closed subsets A, B, C of X ,

$$A \vee B = A \vee C, \text{ but } A' = A \vee (B \cap C) \neq A \vee B.$$

One can enumerate the set $X \setminus A' = \{x_1, \dots, x_n\}$ in such a way that the set $A_i = A' \cup \{x_1, \dots, x_i\}$ is closed for all $i \leq n$. As $B' = B \setminus A'$ and $C' = C \setminus A'$ are nonempty and $B' \cap C' = \emptyset$, there exists $i < n$ such that, say, $B' \subseteq A_i$ but $C' \not\subseteq A_i$, a contradiction. Thus, $\text{Cl}(X, \Phi)$ is join-semidistributive.

To prove the lower semimodularity of $\text{Cl}(X, \Phi)$, observe first that if $A < B$ in $\text{Cl}(X, \Phi)$, then $B = A \cup \{x\}$ for some $x \in B \setminus A$. Indeed, if there exists $x \neq y$ in $B \setminus A$, then $B = \text{Cl}(A \cup \{x\}) = \text{Cl}(A \cup \{y\})$, which contradicts (AEP).

Now if $A < A \vee B$ in $\text{Cl}(X, \Phi)$, then $A \vee B = A \cup \{x\}$ for some $x \in B \setminus A$, hence $A \cap B < B$.

To prove the converse statement, consider a join-semidistributive and lower semimodular lattice L . Denote by $J(L)$ the set of (nonzero) join-irreducible elements and put $J(a) = \{x \in J(L) : x \leq a\}$, for all $a \in L$. Evidently, $J(A) \rightleftharpoons \bigcap \{J(a) : A \subseteq J(a)\}$ defines a closure operator on $J(L)$ and $L \cong \text{Cl}(J(L), J)$.

We prove that $(J(L), J)$ is a finite convex geometry. Consider a nonunit element $a \in L$, and let $b > a$. If $b \in J(L)$, then $J(b) = J(a) \cup \{b\}$. Otherwise the set $C = \{c \in L : c < b, c \not\leq a\}$ is nonempty. By Lemma 1.8, C has a least element c_0 that belongs to $J(L)$. Suppose that C contains another element $d \in J(L)$. Then $a < a \vee d$, hence $a \wedge d < d$ by lower semimodularity, and $a \wedge d = d_*$ is a lower cover of d . As $c_0 < d$ we have also $c_0 \leq d_* \leq a$, a contradiction. Thus, c_0 is the unique element of C , and it is join-irreducible. Hence, $J(b) = J(a) \cup \{c_0\}$ always holds. $\square$

A lattice L is called *atomistic* if any nonzero $a \in L$ is the join of atoms. As any atomistic join-semidistributive lattice is lower semimodular (see Proposition 3.1), we get from Theorem 1.9 the following:

Corollary 1.10. *Any finite atomistic join-semidistributive lattice is the closure lattice of some finite convex geometry.*

We shall discuss in more detail further properties of convex geometries in Section 3, including the connection with combinatorial geometries.

1.3. Universality of finite atomistic convex geometries

Not all finite convex geometries are atomistic (in contrast with combinatorial geometries), see Theorem 1.9.

It turns out that any finite convex geometry can be extended, both join- and meet-preservingly, to some finite atomistic convex geometry defined on the same underlying set.

Theorem 1.11. *Any finite join-semidistributive lattice L can be embedded into the closure lattice of some finite atomistic convex geometry defined on the set $J(L)$ of join-irreducible elements of L . Furthermore, this embedding preserves the bounds.*

By “preserves the bounds” we mean, of course, that a zero and a unit of L is mapped by this embedding to a zero and a unit, respectively, of a convex geometry.

Proof. For any $a \in L$, we put $J(a) = \{x \in J(L) : x \leq a\}$. For all $a, b \in L$ such that $a < b$, it follows from Lemma 1.8 that there exists a least element $p \in L$ such that $p \leq b$ and $p \not\leq a$; we denote this element by $\mu(a, b)$. Furthermore, we put

$$\mathcal{B}(a, b) = \{X \subseteq J(b) \setminus J(a) : \mu(a, b) \in X \Rightarrow X = J(b) \setminus J(a)\}. \quad (1.2)$$

Claim 1. *Let $a, b \in L$ such that $a < b$, let $C \in \mathcal{B}(a, b)$ such that $C \neq J(b) \setminus J(a)$. Then there exists $x \in J(b) \setminus (J(a) \cup C)$ such that $C \cup \{x\} \in \mathcal{B}(a, b)$.*

Proof. From the definition of $\mathcal{B}(a, b)$ follows that $\mu(a, b) \notin C$. We put $x = \mu(a, b)$ if $J(b) \setminus (J(a) \cup C) = \{\mu(a, b)\}$, and we pick any $x \neq \mu(a, b)$ in $J(b) \setminus (J(a) \cup C)$ otherwise. Then x satisfies the required conditions. $\square$

For $x \in L$, we shall construct, by induction on x , a subset T_x of $B(J(L))$:

(T0) We put $T_{0_L} = \{\emptyset\}$.

(T1) For $x \in J(L)$, we put

$$T_x = T_{x_*} \cup \{B \cup \{x\} : B \in T_{x_*}\},$$

where x_* denotes the unique lower cover of x .

(T2) For $x \in L \setminus (J(L) \cup \{0_L\})$, we put

$$T_x = T_{<x} \cup \{J(y) \cup C : y < x, C \in \mathcal{B}(y, x)\},$$

where we put $T_{<x} = \bigcup_{y < x} T_y$ and $\mathcal{B}(y, x)$ is defined as in (1.2).

Hence, all T_x , for $x \in L$, are subsets of $B(J(L))$. We put $T = T_{1_L}$.

Further elementary properties of the sets T_x are summarized in the following claim:

Claim 2. *The following assertions hold:*

- (i) $x \leq y$ implies that $T_x \subseteq T_y$, for all $x, y \in L$;
- (ii) $\{x\} \in T_x$ for all $x \in J(L)$;
- (iii) $J(x)$ is the largest element of T_x for all $x \in L$.

Claim 3. *Let $x, y \in L$ such that $x < y$. Then there exists $z \in J(y) \setminus J(x)$ such that $J(x) \cup \{z\} \in T_y$.*

Proof. We first replace y by any $y' \in L$ such that $x < y' \leq y$, so we see that without loss of generality, we may assume that $x < y$. If $y \in J(L)$, then $x = y_*$, thus, by Claim 2(iii), $J(x) \cup \{y\} = J(y) \in T_y$, so we may take $z = y$. Suppose now that $y \notin J(L)$. By Claim 1, there exists $z \in J(y) \setminus J(x)$ such that $\{z\} \in \mathcal{B}(x, y)$. Then $J(x) \cup \{z\} \in T_y$. $\square$

Claim 4. *For all $B \in T$ with $B \neq J(L)$, there exists $x \notin B$ such that $B \cup \{x\} \in T$.*

Proof. The case where $B = J(a)$ for some $a \in L$ is taken care of by Claim 3. Thus, to conclude the proof of Claim 4, it suffices to prove that for any $x \in L$ and any $B \in T_x \setminus T_{<x}$, if $B \neq J(x)$, then there exists $y \in J(x) \setminus B$ such that $B \cup \{y\} \in T_x$.

We argue by induction on x . The statement holds vacuously if $x = 0_L$. Now suppose that $x > 0_L$, and let $B \in T_x \setminus T_{<x}$ such that $B \neq J(x)$. We consider two cases.

Case 1. $x \notin J(L)$. So $B = J(y) \cup C$ for some $y < x$ and some $C \in \mathcal{B}(y, x)$. From $B \neq J(x)$ follows that $C \neq J(x) \setminus J(y)$. By Claim 1, there exists $z \in J(x) \setminus (J(y) \cup C)$ such that $C \cup \{z\} \in \mathcal{B}(y, x)$. Hence $z \notin B$ and $B \cup \{z\} = J(y) \cup (C \cup \{z\}) \in T_x$.

Case 2. $x \in J(L)$. Then $B = C \cup \{x\}$ for some $C \in T_{x_*}$. Let $y \leq x_*$ be a minimal element of L such that $C \in T_y$. Hence $C \in T_y \setminus T_{<y}$.

- If $C \neq J(y)$, then, by the induction hypothesis, there exists $z \in J(y) \setminus C$ such that $C \cup \{z\} \in T_y$. Therefore, $z \in J(x) \setminus B$ and $B \cup \{z\} = (C \cup \{z\}) \cup \{x\} \in T_x$.
- Finally, suppose that $C = J(y)$. If $y = x_*$, then $B = J(x_*) \cup \{x\} = J(x)$, a contradiction. Hence $y < x_*$. By Claim 3, there exists $z \in J(x_*) \setminus J(y)$ such that $C \cup \{z\} \in T_{x_*}$. So $z \notin B$, and $B \cup \{z\} = (C \cup \{z\}) \cup \{x\}$ belongs to T_x .

This concludes the proof of Claim 4. $\square$

Claim 5. Let $\mathcal{S}(T)$ be the meet-subsemilattice of $\mathbf{B}(\mathbf{J}(L))$ generated by T . Then the closure space on $\mathbf{J}(L)$ induced by $\mathcal{S}(T)$ is a finite atomistic convex geometry.

Proof. The “atomistic” part of the statement follows immediately from Claim 2(ii). Now we verify the “convex geometry” statement. For any $\mathcal{X} \subseteq \mathbf{B}(\mathbf{J}(L))$, we put $\mathcal{X}^{(0)} = \mathcal{X}$, $\mathcal{X}^{(1)} = \{B \cap C : B, C \in \mathcal{X}^{(0)}\}$, and $\mathcal{X}^{(k+1)} = (\mathcal{X}^{(k)})^{(1)}$ for all $k \geq 0$. As $\mathbf{J}(L)$ is finite, $\mathcal{S}(T) = T^{(n)}$ for some n . Hence, in view of Claim 4, it suffices to prove that the assertion “for any $B \in \mathcal{X}$, $B \neq \mathbf{J}(L)$, there exists $x \notin B$ such that $B \cup \{x\} \in \mathcal{X}$ ” can be lifted from $\mathcal{X}$ to $\mathcal{X}^{(1)}$.

Let $\mathcal{X}$ have this property, and let $D = B \cap C$ for some $B, C \in \mathcal{X}$. We may assume without loss of generality that $B \setminus D \neq \emptyset$. By assumption, the set $\mathbf{J}(L) \setminus C = \{y_1, \dots, y_k\}$ can be enumerated in such a way that $C \cup \{y_1, \dots, y_i\} \in \mathcal{X}$ for all $i \in \{1, \dots, k\}$. Observe that $B \setminus D$ is a nonempty subset of $\{y_1, \dots, y_k\}$. Let y_m be the element of $B \setminus D$ with the smallest index. Then $y_1, \dots, y_{m-1} \notin B \setminus D$, hence $B \cap (C \cup \{y_1, \dots, y_m\}) = D \cup \{y_m\} \in \mathcal{X}^{(1)}$, and we are done. $\square$

It remains to prove that L is embedded in $\mathcal{S}(T)$ via the rule $a \mapsto \mathbf{J}(a)$. Evidently, the map $x \mapsto \mathbf{J}(x)$ is a one-to-one meet-homomorphism from L into $\mathcal{S}(T)$.

Let us prove that $\mathbf{J}(x \vee y) = \mathbf{J}(x) + \mathbf{J}(y)$, where $+$ denotes the join in $\mathcal{S}(T)$, for all $x, y \in L$. As any element in $\mathcal{S}(T)$ is a meet of elements from T , it suffices to prove that, for any $B \in T$, if $\mathbf{J}(x) \cup \mathbf{J}(y) \subseteq B$, then $\mathbf{J}(x \vee y) \subseteq B$. The result is trivial if x and y are comparable, so, suppose that x and y are incomparable, in notation, $x \parallel y$. Let $z \in L$ be minimal such that $B \in T_z$. In particular, $B \subseteq \mathbf{J}(z)$, so from $x \parallel y$ follows that $x, y < z$. We now argue by cases.

Case 1: $z \in \mathbf{J}(L)$. Then $\{z\} \subseteq B \subseteq \mathbf{J}(z)$. We argue by induction on $|B|$. From $x, y < z$ follows that $x, y \leq z_*$, thus, by the definition of T_z , $B = C \cup \{z\}$ for some $C \in T_{z_*}$. Therefore, $\mathbf{J}(x) \cup \mathbf{J}(y) \subseteq C$, so, by the induction hypothesis, $\mathbf{J}(x \vee y) \subseteq C \subseteq B$.

Case 2: $z \notin \mathbf{J}(L)$. Then $\mathbf{J}(t) \subseteq B \subseteq \mathbf{J}(z)$ for some $t < z$. If $\mathbf{J}(x) \cup \mathbf{J}(y) \subseteq \mathbf{J}(t)$, then $x \vee y \leq t$, whence $\mathbf{J}(x \vee y) \subseteq \mathbf{J}(t) \subseteq B$. Thus, we may now suppose that $\mathbf{J}(y) \not\subseteq \mathbf{J}(t)$, that is, $y \not\leq t$. Observe that $y \leq z$. Hence, $\mu(t, z) \leq y$, so $\mu(t, z) \in \mathbf{J}(y) \subseteq B$, from which it follows, by the definition of T_z , that $B = \mathbf{J}(z)$, so $\mathbf{J}(x \vee y) \subseteq B$. $\square$

The proof of Theorem 1.11 is completed. $\square$

Remark 1.12. It is well known that the modular analogue of Theorem 1.11 fails. For example, let K be the subgroup lattice of $(\mathbb{Z}/4\mathbb{Z})^3$, so K is modular. Moreover, K is subdirectly irreducible, hence it follows easily from Frink’s embedding Theorem and the results of Herrmann and Huhn [40] that K cannot be embedded into any complemented modular lattice. Hence, a fortiori, it cannot be embedded into any finite atomistic modular lattice.

There is a unique “largest” finite convex geometry extending a given finite convex geometry (X, φ) . We shall call a finite convex geometry (X, ψ_1) an *extension* of a finite convex geometry (X, ψ_0) defined on the same underlying set X , if $\text{Cl}(X, \psi_0)$ is a

sublattice of $\text{Cl}(X, \psi_1)$. The extensions of a given finite convex geometry (X, ψ_0) can be ordered as follows: if (X, ψ_1) and (X, ψ_2) are two such extensions, we say that $(X, \psi_1) \leq (X, \psi_2)$ if $\psi_2(Y) \subseteq \psi_1(Y)$ for any $Y \subseteq X$; equivalently, $\text{Cl}(X, \psi_1) \subseteq \text{Cl}(X, \psi_2)$.

Corollary 1.13. *Any finite convex geometry (X, φ) has a largest extension. Moreover, this extension is atomistic.*

We observe here that the construction of Theorem 1.11 does not give the largest finite convex geometry extending the given lattice L . See further details in Remark 1.15.

Proof. Put $L = \text{Cl}(X, \varphi)$. Since $\varphi(\emptyset) = \emptyset$, φ separates points and satisfies the anti-exchange property, the rule $x \mapsto \varphi(\{x\})$ defines an isomorphism from (X, φ) onto $(J(L), J)$.

Let $(X, \psi_1), \dots, (X, \psi_n)$ be all the different extensions of (X, φ) . It follows from the previous paragraph and from Theorem 1.11 that one of the extensions $(X, \psi_1), \dots, (X, \psi_n)$ is atomistic. Now we define a closure operator ψ on X by the rule $\psi(Y) = \bigcap_{i \leq n} \psi_i(Y)$, for all $Y \subseteq X$. It is proved in [20] that (X, ψ) is a finite convex geometry. It is not hard to prove that (X, ψ) is an extension of (X, φ) as well. Indeed, if $a \vee b = c$ in $\text{Cl}(X, \varphi)$ then $a \vee b = c$ in $\text{Cl}(X, \psi_i)$ for any $i \leq n$, hence $a \vee b = c$ in $\text{Cl}(X, \psi)$. Thus, $\text{Cl}(X, \psi)$ coincides with one of $(X, \psi_1), \dots, (X, \psi_n)$ and $\text{Cl}(X, \psi_i) \subseteq \text{Cl}(X, \psi)$, for all $i \leq n$. Since one of the closure spaces (X, ψ_i) is atomistic, (X, ψ) is atomistic as well. $\square$

From Theorems 1.4 and 1.11, we also get

Corollary 1.14. *The quasivariety $\mathbf{SD}_\vee$ is generated by the closure lattices of finite atomistic convex geometries.*

Remark 1.15. Lower bounded lattices form a proper subclass of join-semidistributive lattices (see definition and further details in Section 3). There is a standard way to embed a finite lattice into a finite atomistic lattice, see [64], and this procedure preserves lower boundedness. We recall here the main features of this construction.

Let L be a finite lattice. A subset $U \subseteq J(L)$ is called a *minimal cover* of an element $v \in J(L)$ if

$$v \leq \bigvee U \text{ and } v \not\leq u_* \vee \bigvee (U \setminus \{u\}) \text{ for all } u \in U.$$

Consider the lattice $\mathbf{A}(L)$ whose elements are the sets $X \subseteq J(L)$ that contain $v \in J(L)$ whenever some minimal cover of v is a subset of X . Clearly, $\mathbf{A}(L)$ is an atomistic lattice with atoms $\{v\}$, $v \in J(L)$. Moreover, the rule $a \mapsto J(a)$ defines a lattice embedding of L into $\mathbf{A}(L)$, see [64].

Let us prove that this embedding does not preserve the join-semidistributive law as a rule. Consider the lattice L depicted in Fig. 1. It is easy to verify that L is

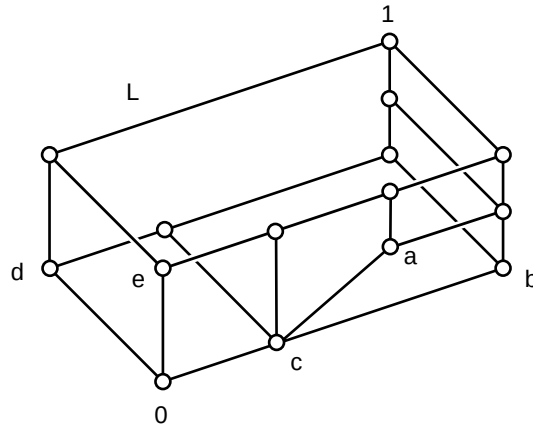


Fig. 1.

join-semidistributive, $J(L) = \{a, b, c, d, e\}$, and $\mathbf{A}(L)$ is defined by the following relations:

$$a \leq b \vee e, \quad b \leq a \vee d, \quad a \vee b \leq c \vee d \vee e.$$

In particular, it follows that

$$a \vee (d \vee e) = b \vee (d \vee e) \neq d \vee e,$$

whence $\mathbf{A}(L)$ is not join-semidistributive.

Remark 1.16. In contrast with the construction presented above, the embedding of a finite join-semidistributive lattice into the atomistic join-semidistributive lattice constructed in Theorem 1.11 does not preserve lower boundedness as a rule.

For a poset P , let $\mathbf{Co}(P)$ denote the lattice of convex subsets of P , as defined in Section 3.1. Let $\mathbf{4}$ denote the four-element chain. Then $\mathbf{Co}(\mathbf{4})$ is the lattice defined by the generators (atoms) a, b, c, d and the relations $b \leq a \vee c$, $c \leq b \vee d$, and $b, c \leq a \vee d$. Let $L = \mathbf{Co}(\mathbf{4}) \setminus \{a, d\}$. Then $L \cong \mathbf{3} \times \mathbf{3}$, and it is easy to verify that the construction of Theorem 1.11 embeds L into $\mathbf{Co}(\mathbf{4})$, the latter known not to be lower bounded. Since the largest extension of L is the Boolean lattice $\mathbf{B}(\mathbf{4})$, we also conclude that the construction of Theorem 1.11 does not provide the largest extension for an arbitrary join-semidistributive lattice.

Still, both the largest extension and the extension provided by the proof of Theorem 1.11 are *intrinsic*, for example, every automorphism of L extends to an automorphism of the extension. This property is a priori not satisfied by the construction given in the proof of Theorem 4.3.3 in [34], which depends of the initial choice of a linear extension of the partial ordering of L .

2. Lattices of algebraic subsets

Let L be a complete lattice. A subset $X \subseteq L$ is called *algebraic*, if X is closed under arbitrary meets and under joins of nonempty up-directed subsets. In particular, $\bigwedge \emptyset = 1_L \in X$. Note that if L is an algebraic lattice, then X , endowed with the induced ordering, is also an algebraic lattice. We shall also need the following classical lemma about algebraic lattices. We say that a complete lattice L is *upper continuous*, if $a \wedge \bigvee_{i \in I} x_i = \bigvee_{i \in I} (a \wedge x_i)$ for any $a \in A$, any up-directed poset I , and any increasing family $(x_i)_{i \in I}$ of elements of A . Let *lower continuous* lattices be defined dually. We say that a lattice L is *weakly atomic*, if for all $a < b$ in A , there are $x, y \in A$ such that $a \leq x < y \leq b$. We say that L is *spatial*, if every element of L is a join of completely join-irreducible elements of L .

Lemma 2.1. *Every algebraic lattice A satisfies the following properties:*

- (i) A is upper continuous.
- (ii) A is weakly atomic.
- (iii) A is dually spatial.

Proof. (i) See [13, Lemma 2.3].

(ii) See [13, Lemma 2.2], or [34, Exercise 1.3.1].

(iii) See [13, Proposition 6.1], or [30, Theorem I.4.22]. Also [34, Lemma 1.3.2]. $\square$

Let $S_p(L)$ denote the lattice of algebraic subsets of L ordered by containment. It is easy to prove that $S_p(L)$ is dually isomorphic to the lattice of continuous closure operators defined on L (see [34, Proposition 1.3.10]). The lattices of algebraic subsets emerged in the paper by Gorbunov and Tumanov [35] where the authors prove that for any algebraic lattice A the lattice $S_p(A)$ is isomorphic to $\mathcal{L}_q(\mathcal{K})$ for some quasivariety $\mathcal{K}$ of relational structures. In particular, $S_p(A)$ is join-semidistributive and dually algebraic. It turns out that the conclusion of join-semidistributivity can be reached under weaker assumptions on the lattice A , see also [33].

Theorem 2.2. *Let A be a complete lattice. Then the following assertions hold:*

- (i) *If A is upper continuous, then $S_p(A)$ is join-semidistributive and lower continuous.*
- (ii) *If A is algebraic, then $S_p(A)$ is dually algebraic.*

Proof. (i) See [34, Theorem 4.1.1].

(ii) See [34, Theorem 5.6.7]. $\square$

For arbitrary complete A , some additional information can be found in Proposition 3.15.

Example 2.3. Let $A = [0, 1]$ be the real unit interval. Then $S_p(A)$ consists of all topologically closed subsets of $[0, 1]$ that contain 1 as an element. Hence, $\{1\}$ is the

only dually compact element of $\mathbf{S}_p(A)$, so $\mathbf{S}_p(A)$ is not dually algebraic, although A is upper continuous.

The principal aim of Section 2 is to prove the following statement.

Theorem 2.4. *Let L be a join-semidistributive lattice defined, within $\mathbf{SD}_\vee$, by finitely many relations. Then there exists an algebraic and dually algebraic lattice A such that L embeds into $\mathbf{S}_p(A)$, via an embedding that preserves the existing bounds.*

Remark 2.5. Let $\mathcal{L}_q(\mathcal{K})$ denote the class of all subquasivarieties of a given quasivariety $\mathcal{K}$. It is proved in [33] that the “algebraic” part of conclusion of Theorem 2.4 also holds for all lattices of the form $\mathcal{L}_q(\mathcal{K})$, that is, there exists a 0, 1-embedding of $\mathcal{L}_q(\mathcal{K})$ into $\mathbf{S}_p(A)$ for some algebraic lattice A .

Remark 2.6. As any free lattice is join-semidistributive (and defined by the empty set of relations!), Theorem 2.4 yields that *any free lattice embeds into $\mathbf{S}_p(A)$ for some algebraic and dually algebraic lattice A .*

Another class of lattices to which Theorem 2.4 applies is the class of all *finitely presented* (within $\mathbf{SD}_\vee$) join-semidistributive lattices, hence *any finitely presented join-semidistributive lattice has a 0, 1-embedding into $\mathbf{S}_p(A)$ for some algebraic and dually algebraic lattice A .*

Together with the result of [35] mentioned above, this gives the following

Corollary 2.7. *The quasivariety $\mathbf{SD}_\vee$ of join-semidistributive lattices is generated by the class of lattices of quasivarieties.*

Observe that the class of lattices of quasivarieties of algebras generates a proper subquasivariety of $\mathbf{SD}_\vee$, see [28,31].

The proof of Theorem 2.4 is based on the next two statements having an interest of their own.

Theorem 2.8. *For any lattice L additively generated by a subset A such that $0 \notin A$, there exists a dually algebraic lattice $\mathbf{T}(L, A)$ such that L is embedded into $\mathbf{S}_p(\mathbf{T}(L, A))$. Furthermore, this embedding preserves the existing bounds.*

Theorem 2.9. *If L is a finite join-semidistributive lattice and A is the set of atoms of L , then $\mathbf{T}(L, A)$ is both algebraic and dually algebraic.*

Remark 2.10. It follows from Theorems 1.11, 2.8 and 2.9 that *every finite join-semidistributive lattice L embeds into $\mathbf{S}_p(A)$ for some algebraic lattice A .* This lattice A is, in general, *infinite*, although L is finite. The fact that A has to be infinite is fully justified by Theorem 3.27, that states that A can be taken finite iff L is *lower bounded*.

This strange phenomenon contrasts with other representation results such as the one of Pudlák and Tůma, see [60], that states that every finite lattice embeds into some *finite* partition lattice.

The construction of the lattice $T(L, A)$ is given in Section 2.1 and the proofs of Theorems 2.8 and 2.9 are given in Sections 2.2 and 2.3, respectively.

Before proving Theorem 2.4, we state a lemma:

Lemma 2.11. *Let $(A_i)_{i \in I}$ be a family of algebraic (and dually algebraic) lattices. Then there exists an algebraic (and dually algebraic) lattice A such that $\prod_{i \in I} S_p(A_i)$ has a 0, 1-embedding into $S_p(A)$.*

Proof (see also Gorbunov [35]). We first observe that the product $A = \prod_{i \in I} A_i$ is an algebraic (and dually algebraic) lattice. We can define a map

$$f: \prod_{i \in I} S_p(A_i) \rightarrow S_p(A)$$

by the rule

$$f((X_i)_{i \in I}) = \prod_{i \in I} X_i \text{ for all } (X_i)_{i \in I} \in \prod_{i \in I} S_p(A_i).$$

It is not hard to verify that f is a 0, 1-embedding of lattices. $\square$

Proof of Theorem 2.4. Let L be a lattice defined within $\mathbf{SD}_\vee$ by finitely many relations. Then, by Theorem 2.4.1 in [34], this lattice is limit-projective in $\mathbf{SD}_\vee$, which, in particular, implies that if L is embedded into a superdirect limit of a direct spectrum of lattices from $\mathbf{SD}_\vee$, then it is embedded into some lattice from that direct spectrum. In view of Theorem 2.3.6 in [34], the quasivariety $\mathbf{Q}(\mathbf{K})$ generated by a class $\mathbf{K}$ coincides with $\mathbf{L}_s \mathbf{P}_s(\mathbf{K})$, where $\mathbf{L}_s$ and $\mathbf{P}_s$ are the operators of taking superdirect limits and subdirect products, respectively. Since the quasivariety $\mathbf{SD}_\vee$ is generated by its finite members (Theorem 1.4), it follows that there exists a subdirect decomposition of L of the form

$$L \hookrightarrow \prod_{i \in I} L_i, \quad \text{for finite join-semidistributive lattices } L_i, i \in I. \quad (2.1)$$

Observe, in particular, that the embedding of (2.1) preserves the existing bounds. In view of Theorem 1.11, Theorems 2.8 and 2.9 imply that any finite join-semidistributive lattice can be 0, 1-embedded into $S_p(A)$ for some algebraic (and dually algebraic) lattice A . Applying this to L_i , for $i \in I$, yields an algebraic (and dually algebraic) lattice A_i such that L_i has a 0, 1-embedding into $S_p(A_i)$. By Lemma 2.11, $\prod_{i \in I} S_p(A_i)$ can be 0, 1-embedded into $S_p(A)$ for a suitable algebraic (and dually algebraic) lattice A . $\square$

More information about the lattices $S_p(A)$ can be found in Section 3.6.

We now turn to the proofs of Theorems 2.8 and 2.9.

2.1. Principal construction

Let $(L, +, \cdot)$ be an arbitrary lattice and let A be a fixed subset in L that additively generates L . Note that until the end of the proof we will be using $+$, $\cdot$ to denote the operations of join and meet on the lattice L in order to distinguish them from those on the lattice $T(L, A)$ that we shall construct.

We shall consider the set $R(A)$ whose elements are the nonempty words $\alpha = \langle a_1, a_2, \dots, a_n \rangle$ on the alphabet A that satisfy the condition

$$a_i \not\leq a_{i+1}, \quad \text{for all } i \leq n-1. \quad (2.2)$$

We shall put $|\alpha| = n$, the *length* of α .

Greek letters will always denote words or segments of words, that is, one or several consecutive letters of a word, while roman font will be kept for letters from A . By $e(\alpha)$ we shall denote the last letter a_n in α and the remainder of the word will be denoted by $\bar{\alpha}$. Thus $\alpha = \bar{\alpha} \frown e(\alpha)$, where $\frown$ stands for *concatenation* of sequences.

We define a partial order $\sqsubseteq$ on $R(A)$ setting $\alpha = \langle a_1, a_2, \dots, a_m \rangle \sqsubseteq \langle b_1, b_2, \dots, b_n \rangle = \beta$ iff $m \leq n$ and there exist $1 \leq j_1 < \dots < j_m \leq n$ such that $a_i = b_{j_i}$ for all $i \in \{1, \dots, m\}$. If $j_m = n$ then we say that β *dilutes* α , in notation $\alpha \sqsubseteq_{\text{dil}} \beta$, and if $\beta = \alpha \frown \gamma$ for some γ then we say that β *extends* α , in notation $\alpha \sqsubseteq_{\text{ext}} \beta$. If $\alpha \sqsubseteq_{\text{ext}} \beta$ and $\alpha \neq \beta$, we write $\alpha \sqsubset_{\text{ext}} \beta$. If $\alpha \sqsubseteq \beta$ and $\alpha \neq \beta$, we write $\alpha \sqsubset \beta$. We extend the symbol $\sqsubset$ to $R(A) \cup \{\emptyset\}$ by stating that $\emptyset \sqsubset \alpha$, for any $\alpha \in R(A)$. It is clear that $\{\beta \in R(A) : \beta \sqsubseteq \alpha\}$ is finite, for each $\alpha \in R(A)$.

For a subset X of $R(A)$, we denote by $\text{Min } X$ the set of minimal elements of X with respect to $\sqsubseteq$. For any $k \geq 1$ we define a *partial* $(k+1)$ -ary operation f_k on $R(A)$ as follows:

$$\alpha = f_k(\alpha_0, \dots, \alpha_k), \quad \text{if } \alpha = \bar{\alpha}_0 = \dots = \bar{\alpha}_k \text{ and } e(\alpha) \leq e(\alpha_0) + \dots + e(\alpha_k).$$

A subset $F \subseteq R(A)$ will be called an *L-filter* in $R(A)$ if it is an order-filter in $(R(A), \sqsubseteq)$ which is closed with respect to the f_k 's, that is, $\alpha \in F$ whenever $k \geq 1$, $\alpha_0, \dots, \alpha_k \in F$, and $\alpha = f_k(\alpha_0, \dots, \alpha_k)$. In particular, both $\emptyset$ and $R(A)$ are L-filters in $R(A)$.

Evidently, the set of all L-filters in $R(A)$ forms a complete lattice $T(L, A) = (T, \vee, \wedge)$ with respect to the *reverse* inclusion.

Lemma 2.12. *For any $\alpha \in R(A)$, the set $R(\alpha) = \{\beta \in R(A) : \alpha \sqsubseteq \beta\}$ is an L-filter.*

Proof. Let $\alpha \sqsubseteq \alpha_0, \dots, \alpha_k$. We have to prove that $\alpha \sqsubseteq f_k(\alpha_0, \dots, \alpha_k)$. So $\bar{\alpha}_0 = \dots = \bar{\alpha}_k$ and $e(\bar{\alpha}_0) \leq e(\alpha_0) + \dots + e(\alpha_k)$. If $e(\alpha) = e(\alpha_j)$ for all $j \leq k$ then $e(\bar{\alpha}_0) \leq e(\alpha_0)$, which contradicts (2.2). Hence $\alpha \sqsubseteq \bar{\alpha}_0$. $\square$

For $X \subseteq R(A)$ and $n \in \omega$, we define a subset $X^{(n)}$ of $R(A)$ by $X^{(0)} = X$ and $X^{(n+1)} = (X^{(n)})^{(1)}$ for any n , with

$$X^{(1)} = X \cup \{\alpha \in R(A) : \exists k \geq 1, \\ \exists \alpha_0, \dots, \alpha_k \in X \text{ such that } f_k(\alpha_0, \dots, \alpha_k) \sqsubseteq \alpha\}.$$

Evidently, the sequence $(X^{(n)})_{n \in \omega}$ is increasing with respect to inclusion. We put $\bar{X} = \bigcup_{n \in \omega} X^{(n)}$. Observe the following obvious fact.

Lemma 2.13. *For any $X \subseteq R(A)$, the set $\bar{X}$ is the smallest L -filter in $R(A)$ containing X . In particular, for any family $(F_i)_{i \in I}$ in $\mathbf{T}(L, A)$, we get $\bigwedge_{i \in I} F_i = \overline{\bigcup_{i \in I} F_i}$.*

It is obvious that $X^{(1)} = \bigcup \{Y^{(1)} : Y \subseteq X, |Y| < \omega\}$ for any $X \subseteq R(A)$. We get by induction that the analogous equality holds for $X^{(n)}$, for any n . Hence, the closure operator $X \mapsto \bar{X}$ on $R(A)$ is algebraic, which implies the following observation:

Lemma 2.14. *The lattice $\mathbf{T}(L, A)$ is dually algebraic.*

2.2. Proof of Theorem 2.8

In this subsection, we assume that L is a lattice, additively generated by a subset A of L , and that $0 \notin A$.

Recall that by $\text{Min } F$ we denote the set of minimal elements of a given subset $F \subseteq R(A)$. For any $a \in L$, we set

$$\psi(a) = \{F \in \mathbf{T}(L, A) : e(\alpha) \leq a \text{ for all } \alpha \in \text{Min } F\}.$$

It is obvious that if L has a zero, then $\psi(0_L) = \{\emptyset\}$ is the zero of $\mathbf{S}_p(\mathbf{T}(L, A))$, while if L has a unit, then $\psi(1_L) = \mathbf{T}(L, A)$ is the unit of $\mathbf{S}_p(\mathbf{T}(L, A))$. Hence, ψ preserves the existing bounds.

Lemma 2.15. *The set $\psi(a)$ is an algebraic subset of $\mathbf{T}(L, A)$.*

Proof. Let $\{F_i : i \in I\}$ be an arbitrary up-directed subset in $\psi(a)$. We shall prove that $F = \bigvee_{i \in I} F_i = \bigcap_{i \in I} F_i$ belongs to $\psi(a)$. Let α in $\text{Min } F$. There are only finitely many elements less than α in the poset $(R(A), \sqsubseteq)$, hence α is minimal in F_j for some $j \in I$. This yields that $e(\alpha) \leq a$.

Now consider an arbitrary family $\{F_i : i \in I\}$ of filters in $\psi(a)$. We have to check that $F = \bigwedge_{i \in I} F_i$ belongs to $\psi(a)$. As $F = \overline{\bigcup_{i \in I} F_i}$, it is enough to prove that $e(\alpha) \leq a$ holds for any minimal element α of $(\bigcup_{i \in I} F_i)^{(n)}$, for any $n \in \omega$. We proceed by induction on n .

For $n = 0$ we have $e(\alpha) \leq a$ by assumption. Suppose that the statement holds for n and let α be a minimal element of $(\bigcup_{i \in I} F_i)^{(n+1)}$. We may assume that $\alpha \notin (\bigcup_{i \in I} F_i)^{(n)}$. By definition, $\alpha \sqsubseteq f_k(\beta_0, \dots, \beta_k)$ for some $k \geq 1$ and $\beta_0, \dots, \beta_k \in (\bigcup_{i \in I} F_i)^{(n)}$. It follows from the minimality of α that $\alpha = f_k(\beta_0, \dots, \beta_k)$. By assumption, $\alpha \neq \beta_0$. Hence, $\alpha = \overline{\beta_0} = \dots = \overline{\beta_k}$ and $e(\alpha) \leq e(\beta_0) + \dots + e(\beta_k)$. It remains to prove that $e(\beta_j) \leq a$ for all $j \leq k$. For all $j \in \{0, \dots, k\}$, there exists a minimal element γ_j of $(\bigcup_{i \in I} F_i)^{(n)}$ such that $\gamma_j \sqsubseteq \beta_j$. If $e(\beta_j) \neq e(\gamma_j)$ then $\gamma_j \sqsubseteq \overline{\beta_j} = \alpha$. This implies that $\gamma_j = \alpha$ as α is a minimal element of $(\bigcup_{i \in I} F_i)^{(n+1)}$, and we get a contradiction with the assumption $\alpha \notin (\bigcup_{i \in I} F_i)^{(n)}$. Consequently, $e(\beta_j) = e(\gamma_j)$, so, by the induction hypothesis, $e(\gamma_j) \leq a$. Hence $e(\alpha) \leq a$. $\square$

Lemma 2.16. *The map ψ is an isomorphic embedding of L into $\mathbf{S}_p(\mathbf{T}(L, A))$.*

Proof. Evidently, ψ preserves the meets. It follows that $\psi(a) \vee \psi(b) \subseteq \psi(a + b)$. Now we prove the converse inclusion.

Consider $F \in \psi(a + b)$. By Lemma 2.12,

$$F = \bigcup \{R(\gamma) : \gamma \in \text{Min } F\} = \bigwedge \{R(\gamma) : \gamma \in \text{Min } F\},$$

where $\text{Min } F$ is the set of minimal elements of F .

Thus it is enough to prove $R(\gamma) \in \psi(a) \vee \psi(b)$ for any $\gamma \in R(A)$ with $e(\gamma) \leq a + b$.

If $e(\gamma) \leq a$ or $e(\gamma) \leq b$, then $R(\gamma) \in \psi(a)$ or $R(\gamma) \in \psi(b)$, respectively, and we are done. Now suppose that $e(\gamma) \not\leq a, b$. Express $a = x_0 + \dots + x_k$ and $b = y_0 + \dots + y_s$ as joins of elements from A , so, in particular, $e(\gamma) \not\leq x_i$ and $e(\gamma) \not\leq y_j$ for all i, j . Then $\gamma \wedge \langle x_i \rangle$ and $\gamma \wedge \langle y_j \rangle$ belong to $R(A)$, and

$$\gamma = f_{k+s+1}(\gamma \wedge \langle x_0 \rangle, \dots, \gamma \wedge \langle x_k \rangle, \gamma \wedge \langle y_0 \rangle, \dots, \gamma \wedge \langle y_s \rangle)$$

since $e(\gamma) \leq a + b = x_0 + \dots + x_k + y_0 + \dots + y_s$. It follows that

$$R(\gamma) = \bigwedge (R(\gamma \wedge \langle x_i \rangle) : i = 1, \dots, k) \wedge \bigwedge (R(\gamma \wedge \langle y_j \rangle) : j = 1, \dots, s).$$

Clearly, $R(\gamma \wedge \langle x_i \rangle) \in \psi(a) \subseteq \psi(a) \vee \psi(b)$ and $R(\gamma \wedge \langle y_j \rangle) \in \psi(b) \subseteq \psi(a) \vee \psi(b)$, whence $R(\gamma) \in \psi(a) \vee \psi(b)$.

Finally, ψ is one to one. Indeed, if $a \not\leq b$ and $a = x_0 + \dots + x_k$ with $x_i \in A$ then $x_i \not\leq b$ for some $i \leq k$. Put $\alpha = \langle x_i \rangle$. Then $R(\alpha) \in \psi(a)$ and $R(\alpha) \notin \psi(b)$. $\square$

2.3. Proof of Theorem 2.9

We shall start with general considerations that involve neither finiteness nor join-semidistributivity of L .

2.3.1. Minimal elements of X^* and inferences

In this subsection, let L be a lattice, let $A \subseteq L$.

For $X \subseteq R(A)$ and $n \in \omega$, we define a subset $X^{[n]}$ of $R(A)$ by $X^{[0]} = X$ and $X^{[n+1]} = (X^{[n]})^{[1]}$ for any n , and the following

$$X^{[1]} = X \cup \{\alpha \in R(A) : \exists k \geq 1, \\ \exists \alpha_0, \dots, \alpha_k \in X \text{ such that } f_k(\alpha_0, \dots, \alpha_k) = \alpha\}.$$

Evidently, the sequence $(X^{[n]})_{n \in \omega}$ is increasing with respect to inclusion. We put $X^* = \bigcup_{n \in \omega} X^{[n]}$; hence $X^* \subseteq \bar{X}$. The following two lemmas will relate more precisely the subsets X^* and $\bar{X}$, for any $X \subseteq R(A)$.

It is trivial that $X^{(n)}$ is an order-filter of $(R(A), \sqsubseteq)$, for all $X \subseteq R(A)$ and all $n \geq 1$. The following lemma is slightly less obvious:

Lemma 2.17. *For any order-filter X of $(R(A), \sqsubseteq_{\text{dil}})$ and any $n \in \omega$, the subset $X^{[n]}$ is an order-filter of $(R(A), \sqsubseteq_{\text{dil}})$.*

Proof. It suffices to prove the result for $n = 1$. Let $\alpha \in X^{[1]}$ and $\beta \in R(A)$ such that $\alpha \sqsubseteq_{\text{dil}} \beta$, we prove that $\beta \in X^{[1]}$. If $\alpha \in X$, then this follows from the assumption on X . If $\alpha \notin X$, then we can write $\alpha = f_k(\alpha_0, \dots, \alpha_k)$ for some $k \geq 1$ and $\alpha_0, \dots, \alpha_k \in X$. Put $\beta_i = \beta \frown e(\alpha_i)$, for all $i \in \{0, \dots, k\}$. From $\alpha \sqsubseteq_{\text{dil}} \beta$ follows that $\beta = f_k(\beta_0, \dots, \beta_k)$ and that $\alpha_i \sqsubseteq_{\text{dil}} \beta_i$ for all i , so $\beta_i \in X$ by assumption on X . Therefore, $\beta \in X^{[1]}$. $\square$

Lemma 2.18. *Let X be an order-filter of $(R(A), \sqsubseteq)$. Then $\text{Min } \bar{X} \subseteq X^*$.*

Proof. It is sufficient to prove that $\text{Min } X^{(n)} \subseteq X^{[n]}$, for all $n \in \omega$. We argue by induction on n . The result is trivial for $n = 0$. Suppose that it holds for n , let $\alpha \in \text{Min } X^{(n+1)}$, we prove that $\alpha \in X^{[n+1]}$. If $\alpha \in X^{(n)}$, then $\alpha \in X^{[n]}$ by the induction hypothesis, so $\alpha \in X^{[n+1]}$. Now suppose that $\alpha \notin X^{(n)}$. Then, by the minimality of α , we can write $\alpha = f_k(\alpha_0, \dots, \alpha_k)$, for some $k \geq 1$ and $\alpha_0, \dots, \alpha_k \in X^{(n)}$.

Hence, to conclude the proof, it suffices to prove that $\alpha_i \in X^{[n]}$ for all $i \in \{0, \dots, k\}$. By the induction hypothesis, this holds if $\alpha_i \in \text{Min } X^{(n)}$. If $\alpha_i \notin \text{Min } X^{(n)}$, then there exists $\beta \in \text{Min } X^{(n)}$ such that $\beta \sqsubset \alpha_i$, so, by the induction hypothesis, $\beta \in X^{[n]}$. If $\beta \sqsubseteq \alpha$, then $\alpha \in X^{(n)}$, a contradiction. Hence, $\beta \not\sqsubseteq \alpha = \overline{\alpha_i}$, so it follows from $\beta \sqsubseteq \alpha_i$ that $\beta \sqsubseteq_{\text{dil}} \alpha_i$, whence, since $\beta \in X^{[n]}$ and by Lemma 2.17, $\alpha_i \in X^{[n]}$ again. $\square$

We shall now turn to the most important concept used in the proof of Theorem 2.9. To prepare for it, we define a sub-function f_k^* of f_k , for all $k \geq 1$.

Definition 2.19. For any $k \geq 1$, we define a $(k+1)$ -ary partial operation f_k^* on $R(A)^{k+1}$ by the rule

$$\alpha = f_k^*(\alpha_0, \dots, \alpha_k), \text{ if } \alpha = f_k(\alpha_0, \dots, \alpha_k)$$

and

$$e(\alpha) \not\leq \sum_{i \neq j} e(\alpha_i), \text{ for all } j \in \{0, \dots, k\}.$$

In particular, $\alpha = f_k^*(\alpha_0, \dots, \alpha_k)$ implies that all the $e(\alpha_i)$, for $i \in \{0, \dots, k\}$, are distinct.

Definition 2.20. For a subset X of $R(A)$, let *inferences* from X be the nonempty, finite sequences of elements of $R(A)$ built up according to the following rules.

- (I0) $\langle \alpha \rangle$ is an inference of α from X , for any $\alpha \in X$.
- (I1) If $\alpha \in R(A) \setminus X$, $k \geq 1$, $\alpha_0, \dots, \alpha_k \in R(A)$, $I_0, \dots, I_k$ are inferences of $\alpha_0, \dots, \alpha_k$, respectively, from X , and $\alpha = f_k^*(\alpha_0, \dots, \alpha_k)$, then the finite sequence

$$I = I_0 \frown I_1 \frown \dots \frown I_k \frown \langle \alpha \rangle \quad (2.3)$$

is an inference of α from X .

We shall call expression (2.3) a *canonical decomposition* of I , and we shall call α the *top* of I . As we shall see later (see Corollary 2.25), such a decomposition is *unique*.

If $I = \langle \alpha_0, \dots, \alpha_{n-1} \rangle$, we shall denote by $|I| = n$ the *length* of I , and by $\text{rng } I = \{\alpha_0, \dots, \alpha_{n-1}\}$ the *range* of I .

The following straightforward lemma relates inferences and the correspondence $X \mapsto X^*$.

Lemma 2.21. *Let $X \subseteq R(A)$. Then the following assertions hold:*

- (i) *For any inference I from X , $\text{rng } I$ is a subset of X^* .*
- (ii) *Let $\alpha \in X^*$. Then there exists an inference I of α from X .*

We shall now establish various structural properties of inferences.

Lemma 2.22. *Let $X \subseteq R(A)$, let $\alpha \in R(A)$, let I be an inference of α from X . Then the following assertions hold:*

- (i) *All elements of $\text{rng } I \setminus \{\alpha\}$ are proper extensions of α .*
- (ii) *All entries of I are distinct (i.e., I is a one-to-one sequence).*

Proof. Assertion (i) follows from an obvious induction argument on $|I|$.

(ii) We argue by induction on $|I|$. The result is trivial for $|I| = 1$. If $|I| \geq 2$, we pick a canonical decomposition of I as in (2.3). Let β (resp., γ) be the i -th entry (resp., the j -th entry) of I , with $i \neq j$. If β is the top of I , then, by (i), γ is a proper extension of β . If β, γ belong to the range of the same I_s , with $0 \leq s \leq k$, then $\beta \neq \gamma$ by the induction hypothesis. Suppose that β (resp., γ) belongs to the range of I_r (resp., I_s) with $r \neq s$.

Then, by (i), β (resp., γ) is an extension of α_r (resp. α_s), however, $\alpha_r \neq \alpha_s$ by the definition of f_k^* , hence $\beta \neq \gamma$. $\square$

It follows from Lemma 2.22(ii) that every word of $R(A)$ occurs at most once in an inference I from X . If $\alpha \in \text{rng } I$, then we say that α is a *free word* of I , if $\alpha \in X$, and an *inner word*, otherwise.

Definition 2.23. Let $X \subseteq R(A)$, let I and J be inferences from X . We say that I is a *subinference* of J , in notation $I \triangleleft J$, if $J = I_0 \frown I \frown I_1$ for some finite sequences I_0 and I_1 .

If $I_1 = \emptyset$, we say that I is a *final segment* of J .

If $\text{rng } I \cap \text{rng } J = \emptyset$, we say that I and J are *disjoint*, in notation $I \perp J$.

It follows from Lemma 2.22(ii) that if $I \triangleleft J$, then the finite sequences I_0 and I_1 such that $J = I_0 \frown I \frown I_1$ are *unique*.

Lemma 2.24. Let $X \subseteq R(A)$, let I and J be inferences from X . If I is a final segment of J , then $I = J$.

Proof. We say that I and J are *comparable*, if one is a final segment of the other. We argue by induction on $|J|$. The result is trivial for $|J| = 1$. Suppose now that $|J| \geq 2$ and let $J = J_0 \frown \dots \frown J_n \langle \gamma \rangle$ be a canonical decomposition of J , with J_s an inference of β_s from X , for all $s \in \{0, \dots, n\}$. From the definition of an inference follows that $\gamma \notin X$, whence $I \neq \langle \gamma \rangle$, so I has a canonical decomposition $I = I_0 \frown \dots \frown I_m \langle \gamma \rangle$, with I_r an inference of α_r from X , for all $r \in \{0, \dots, m\}$. Since I and J are comparable, I_m and J_n are comparable, whence, by the induction hypothesis, $I_m = J_n$. An easy induction argument yields similarly that $I_{m-k} = J_{s-k}$ for all $k \in \{0, \dots, \min\{m, n\}\}$. Since I is a final segment of J , $m \leq n$ and J can be decomposed as

$$J = J_0 \frown \dots \frown J_{n-m-1} \frown I_0 \frown \dots \frown I_m \langle \gamma \rangle.$$

Hence, $\gamma = f_n^*(\beta_0, \dots, \beta_{n-m-1}, \alpha_0, \dots, \alpha_m) = f_m^*(\alpha_0, \dots, \alpha_m)$, whence (see Definition 2.19) $m = n$. Therefore, $I = J$. $\square$

As easy consequences of Lemma 2.24, we record the two following corollaries:

Corollary 2.25. Let $X \subseteq R(A)$, let I be an inference from X . Then the following assertions hold:

- (i) For every $\alpha \in \text{rng } I$, there exists a unique subinference J of I with top α .

- (ii) If $|I| \geq 2$ and $I = I_0 \frown \dots \frown I_k \frown \langle \gamma \rangle$ is a canonical decomposition of I , then, for any subinference J of I , either $J = I$, or $J \triangleleft I_s$ for some $s \in \{0, \dots, k\}$.
- (iii) If $|I| \geq 2$, then I has exactly one canonical decomposition.

If $\alpha \in \text{rng } I$, we shall denote by $I(\alpha)$ the unique subinference of I with top α . Observe that α is a free word iff $I(\alpha) = \langle \alpha \rangle$.

Observe also that the definition of $I(\alpha)$ is relative not only to I and α , but also to X . However, X will always be understood from the context.

Corollary 2.26. *Let $X \subseteq R(A)$, let I be an inference from X . Then the subinferences of I from X are nested, that is, for all subinferences J and K of I , either $J \triangleleft K$, or $K \triangleleft J$, or $J \perp K$.*

Lemma 2.27. *Let $X \subseteq R(A)$, let I be an inference from X . For any $\alpha, \beta \in \text{rng } I$, the following are equivalent:*

- (i) $I(\beta) \triangleleft I(\alpha)$;
- (ii) $\text{rng } I(\beta) \subseteq \text{rng } I(\alpha)$;
- (iii) $\alpha \sqsubseteq_{\text{ext}} \beta$.

Proof. Implication (i) $\Rightarrow$ (ii): is trivial.

(ii) $\Rightarrow$ (iii): Assume that (ii) holds. Then $\beta \in \text{rng } I(\beta) \subseteq \text{rng } I(\alpha)$, whence, since $I(\alpha)$ is an inference of α from X and by Lemma 2.22(i), $\alpha \sqsubseteq_{\text{ext}} \beta$.

(iii) $\Rightarrow$ (i): Assume that $\alpha \sqsubseteq_{\text{ext}} \beta$, we prove that $I(\beta) \triangleleft I(\alpha)$, by induction on $|I|$. The conclusion is trivial if $|I| = 1$. If $|I| \geq 2$, then let $I = I(\gamma_0) \frown \dots \frown I(\gamma_k) \frown \langle \gamma \rangle$ be the canonical decomposition of I . If $\alpha = \gamma$, then $I(\alpha) = I$ and the conclusion is trivial. If $\beta = \gamma$, then, by Lemma 2.22(i), $\alpha = \beta$ and the conclusion follows. Now suppose that both α, β are distinct from γ . By Corollary 2.25(ii), there are $r, s \in \{0, \dots, k\}$ such that $I(\alpha) \triangleleft I(\gamma_r)$ and $I(\beta) \triangleleft I(\gamma_s)$. If $r = s$, then $I(\beta) \triangleleft I(\alpha)$ by the induction hypothesis. Suppose that $r \neq s$. By Lemma 2.22(i), $\gamma_s \sqsubseteq_{\text{ext}} \beta$ and $\gamma_r \sqsubseteq_{\text{ext}} \alpha \sqsubseteq_{\text{ext}} \beta$, thus one of γ_r and γ_s is an extension of the other, hence, since they have the same length, $\gamma_r = \gamma_s$, a contradiction. $\square$

Definition 2.28. Let $X \subseteq R(A)$, let I be an inference from X , let $n \in \omega$, let $\alpha_1, \dots, \alpha_n \in \text{rng } I$ such that the subinferences $I(\alpha_1), \dots, I(\alpha_n)$ are pairwise disjoint. We can decompose I as

$$I = I_0 \frown I(\alpha_{\sigma(1)}) \frown I_1 \frown \dots \frown I_{n-1} \frown I(\alpha_{\sigma(n)}) \frown I_n$$

for finite sequences $I_0, \dots, I_n$ and a unique permutation σ of $\{1, \dots, n\}$.

We define a new finite sequence $I[\alpha_1, \dots, \alpha_n]$ by

$$I[\alpha_1, \dots, \alpha_n] = I_0 \frown \langle \alpha_{\sigma(1)} \rangle \frown I_1 \frown \dots \frown I_{n-1} \frown \langle \alpha_{\sigma(n)} \rangle \frown I_n.$$

Observe that the definition of $I[\alpha_1, \dots, \alpha_n]$ is relative not only to I and $\alpha_1, \dots, \alpha_n$, but also to X . However, X will always be understood from the context.

Lemma 2.29. *Let $X \subseteq R(A)$, let I be an inference from X , let $n \in \omega$, and let $\alpha_1, \dots, \alpha_n \in \text{rng } I$ such that $I(\alpha_1), \dots, I(\alpha_n)$ are pairwise disjoint. Then $I[\alpha_1, \dots, \alpha_n]$ is an inference from $X \cup \{\alpha_1, \dots, \alpha_n\}$.*

Proof. It is sufficient to prove the result for $n = 1$. Hence, let $I = J_0 \frown I(\alpha) \frown J_1$, we prove that $I[\alpha] = J_0 \frown \langle \alpha \rangle \frown J_1$ is an inference from $X \cup \{\alpha\}$. We may assume that α is an inner word of I . If $J_1 = \emptyset$ then $I[\alpha] = \langle \alpha \rangle$ and we are done, so suppose that $J_1 \neq \emptyset$. Let $I = I(\beta_0) \frown \dots \frown I(\beta_k) \frown \langle \beta \rangle$ be the canonical decomposition of I . By Corollary 2.25(ii), there exists $s \in \{0, \dots, k\}$ such that $I(\alpha) \triangleleft I(\beta_s)$. By the induction hypothesis, $I(\beta_s)[\alpha]$ is an inference from $X \cup \{\alpha\}$. By Lemma 2.22(ii), all $I(\beta_l)$, for $l \neq s$ in $\{0, \dots, k\}$, are inferences from $X \cup \{\alpha\}$. Therefore,

$$I[\alpha] = I(\beta_0) \frown \dots \frown I(\beta_{s-1}) \frown I(\beta_s)[\alpha] \frown I(\beta_{s+1}) \frown \dots \frown I(\beta_k) \frown \langle \beta \rangle$$

is an inference from $X \cup \{\alpha\}$. $\square$

We now introduce a useful notation, inspired from the intuitive meaning of an inference. For $X \subseteq R(A)$, an inference I from X , $k \geq 1$, $\alpha, \alpha_0, \dots, \alpha_k \in \text{rng } I$, let

$$I \vdash \alpha = f_k^*(\alpha_0, \dots, \alpha_k) \quad (2.4)$$

be the statement that there exists a permutation σ of $\{0, \dots, k\}$ such that

$$I(\alpha) = I(\alpha_{\sigma(0)}) \frown \dots \frown I(\alpha_{\sigma(k)}) \frown \langle \alpha \rangle. \quad (2.5)$$

Definition 2.30. Let $X \subseteq R(A)$, let I be an inference from X , let $\beta, \gamma \in \text{rng } I$. An *I-inference chain* from β to γ is a collection α of finite sequences of the form

$$\alpha = \langle \langle \gamma_{ij} : 0 \leq j \leq k_i \rangle : 0 \leq i \leq n \rangle$$

for some $n \in \omega$, $k_n = 0$, and $k_0, \dots, k_{n-1} \geq 2$, such that the following relations hold:

- (i) $\beta = \gamma_{0,0}$ and $\gamma = \gamma_{n,0}$;
- (ii) $I \vdash \gamma_{i+1,0} = f_{k_i}^*(\gamma_{i,0}, \dots, \gamma_{i,k_i})$ for all $i < n$.

If $\gamma_{i+1,0} = f_{k_i}^*(\gamma_{i,0}, \dots, \gamma_{i,k_i})$, then $\gamma_{i+1,0} = \overline{\gamma_{i,0}}$. Hence, the following Lemma 2.31 is a somewhat more precise version of Lemma 2.22(i).

Lemma 2.31. *Let $X \subseteq R(A)$, let $\gamma \in R(A)$, let I be an inference of γ from X , let $\beta \in \text{rng } I$. Then there exists an I -inference chain from β to γ .*

Proof. We argue by induction on $|I|$. The conclusion is trivial if $|I| = 1$ or $\beta = \gamma$, so suppose that $|I| \geq 2$ and $\beta \neq \gamma$. Let

$$I = I(\gamma_0) \frown \dots \frown I(\gamma_k) \frown \langle \gamma \rangle \quad (2.6)$$

be the canonical decomposition of I . There exists $s \in \{0, \dots, k\}$ such that $\beta \in \text{rng } I(\gamma_s)$. Let σ be any permutation of $\{0, \dots, k\}$ such that $\sigma(0) = s$, and put $\delta_i = \gamma_{\sigma(i)}$, for all $i \in \{0, \dots, k\}$. Then $I \vdash \gamma = f_k^*(\delta_0, \dots, \delta_k)$ with $\delta_0 = \gamma_s$, which gives an I -inference chain from γ_s to γ . Furthermore, by the induction hypothesis (applied to $I(\gamma_s)$), there exists an $I(\gamma_s)$ -inference chain from β to γ_s . The juxtaposition of these two chains gives an I -inference chain from β to γ . $\square$

Corollary 2.32. *Let $X \subseteq R(A)$, let I be an inference from X . Then the set A of inner words of I is convex for the relation $\sqsubseteq_{\text{ext}}$, that is, $\alpha \sqsubseteq_{\text{ext}} \gamma \sqsubseteq_{\text{ext}} \beta$ and $\alpha, \beta \in A$ implies that $\gamma \in A$.*

Proof. It follows from Lemma 2.27 that β and γ belong to $\text{rng } I(\alpha)$. Hence, by working inside $I(\alpha)$, we may suppose that I is an inference of α . By Lemma 2.31, there exists an I -inference chain from β to α , say,

- (i) $\beta = \gamma_{0,0}$ and $\alpha = \gamma_{n,0}$;
- (ii) $I \vdash \gamma_{i+1,0} = f_{k_i}^*(\gamma_{i,0}, \dots, \gamma_{i,k_i})$ for all $i < n$.

In particular, $\gamma_{i+1,0} = \overline{\gamma_{i,0}}$ for all $i < n$, and all $\gamma_{i,0}$, for $0 \leq i \leq n$, are inner words of I . From $\alpha \sqsubseteq_{\text{ext}} \gamma \sqsubseteq_{\text{ext}} \beta$ follows that there exists $i \in \{0, \dots, n\}$ such that $\gamma = \gamma_{i,0}$. $\square$

The following fundamental lemma makes it possible to use X^* instead of $\bar{X}$ in the proofs of most results in Section 2.3.2.

Lemma 2.33. *Let $X \subseteq R(A)$, let $\gamma \in R(A) \setminus X$, let $\beta \in R(A)$ such that $R(\beta) \subseteq X$, let I be an inference of γ from X . Then $\beta \sqsubseteq \delta$ implies that $\beta \sqsubseteq_{\text{dil}} \delta$, for all $\delta \in \text{rng } I$.*

Proof. We argue by induction on $|I|$. Since $\gamma \notin X$, $|I| \geq 2$, so let

$$I = I(\gamma_0) \frown \dots \frown I(\gamma_k) \frown \langle \gamma \rangle$$

be the canonical decomposition of I . Let $\delta \in \text{rng } I$ such that $\beta \sqsubseteq \delta$. It follows from the assumption on X that $\delta \neq \gamma$, so $\delta \in \text{rng } I(\gamma_s)$ for some $s \in \{0, \dots, k\}$. If $\gamma_s \in X$, then $I(\gamma_s) = \langle \delta \rangle$, hence, since $\beta \sqsubseteq \delta$ and $\beta \not\sqsubseteq \gamma = \overline{\gamma_s} = \bar{\delta}$, $\beta \sqsubseteq_{\text{dil}} \delta$. If $\gamma_s \notin X$, then, by the induction hypothesis (applied to $I(\gamma_s)$), $\beta \sqsubseteq_{\text{dil}} \delta$ again. $\square$

2.3.2. $\mathbf{T}(L, A)$ for L finite atomistic

We have already proved in Theorem 2.8 that $\mathbf{T}(L, A)$ is dually algebraic, for any arbitrary lattice L additively generated by a subset A . To prove that $\mathbf{T}(L, A)$ is also

algebraic if L is a finite join-semidistributive atomistic lattice with the set of atoms A , it is enough to prove that $\mathbf{T}(L, A)$ is upper continuous, see Section 2. Indeed, as $\mathbf{T}(L, A)$ is dually algebraic, it follows from Lemma 2.1(iii) that any of its elements is a join of completely join-irreducible elements, and the latter are compact in the presence of upper continuity.

From now on until the end of the proof of Theorem 2.9, we shall fix a finite, join-semidistributive, atomistic lattice L , with set of atoms A , and we shall put $p = |A|$. Towards a contradiction, we shall suppose that $\mathbf{T}(L, A)$ is not upper continuous.

Lemma 2.34. *There exist a filter $F \in \mathbf{T}(L, A)$ and words $\beta, \gamma \in R(A)$ such that $|\beta| \geq |\gamma| + p$ and*

$$\gamma \in (F \wedge R(\beta)) \setminus F.$$

Proof. Let upper continuity fail at some $X \in \mathbf{T}(L, A)$ and chain $\{F_i : i \in I\} \subseteq \mathbf{T}(L, A)$, that is, $X \wedge \bigvee_{i \in I} F_i$ is properly contained in $\bigvee_{i \in I} (X \wedge F_i)$. Then there exists γ that lies in $X \wedge F_i$ for arbitrary $i \in I$ but not in $X_0 = X \wedge \bigvee_{i \in I} F_i$.

For any $i \in I$, put $G_i = F_i \setminus X_0$. Then $X \wedge F_i = \overline{X_0 \cup G_i}$, the least L-filter containing $X_0 \cup G_i$.

Thus, $\gamma \in \overline{X_0 \cup G_i}$, for any $i \in I$, and $\gamma \notin X_0$. Now we want to find G_i all whose words are long enough. More specifically, we need i_0 such that all words in G_{i_0} have length at least $|\gamma| + p$. Such an i_0 exists as $\bigcap_{i \in I} G_i = \emptyset$ and there are only finitely many words whose length is at most $|\gamma| + p - 1$.

Observe that $\gamma \in \overline{X_0 \cup G_{i_0}} \setminus X_0$. Choose the least integer $k \geq 1$ for which there are $\alpha_1, \alpha_2, \dots, \alpha_k \in G_{i_0}$ such that

$$\gamma \in X_0 \wedge R(\alpha_1) \wedge \dots \wedge R(\alpha_k).$$

Now if we denote $F = X_0 \wedge R(\alpha_1) \wedge \dots \wedge R(\alpha_{k-1})$ and $\beta = \alpha_k$ then

$$\gamma \in (F \wedge R(\beta)) \setminus F$$

and $|\beta| \geq |\gamma| + p$. $\square$

Lemma 2.35. *There exist a filter $F \in \mathbf{T}(L, A)$ and words $\beta, \gamma \in R(A)$ such that $\gamma \in (F \cup R(\beta))^* \setminus (F \cup R(\beta))$ and $e(\beta) = e(\gamma)$.*

Proof. Let β, γ , and F be as in the conclusion of Lemma 2.34. Furthermore, by possibly replacing γ by some $\gamma' \sqsubseteq \gamma$, we may assume without loss of generality that $\gamma \in \text{Min}(F \wedge R(\beta))$. Hence, by Lemma 2.18, $\gamma \in (F \cup R(\beta))^*$, thus, by Lemma 2.21(ii), there exists an inference I of γ from $F \cup R(\beta)$.

If all free words of I belong to F , then, by Lemma 2.21(i) and since F is an L-filter, $\gamma \in F$, a contradiction. Hence, there exists a free word β' of I that belongs to $R(\beta) \setminus F$. By Lemma 2.31, there exists an I -inference chain from

β' to γ , say,

- (i) $\beta' = \gamma_{0,0}$ and $\gamma = \gamma_{n,0}$;
- (ii) $I \vdash \gamma_{j+1,0} = f_{k_j}^*(\gamma_{j,0}, \dots, \gamma_{j,k_j})$, for all $j < n$.

It follows from (ii) that $|\gamma_{j,0}| = |\gamma_{j+1,0}| + 1$ for all $j < n$.

Next, we define inductively $\beta_0 = \beta$ and $\beta_{i+1} = \overline{\beta_i}$, for all $i < p$. There are indices $s < r \leq p$ such that $e(\beta_r) = e(\beta_s)$. Since $\beta_r \sqsubset_{\text{ext}} \beta \sqsubseteq \beta'$, there exists $\gamma' \sqsubset_{\text{ext}} \beta'$ such that $\beta_r \sqsubseteq_{\text{dil}} \gamma'$; we choose γ' with minimal length. We observe that $|\gamma'| \geq |\beta_r| = |\beta| - r \geq |\beta| - p \geq |\gamma|$, hence (since β' extends both γ and γ') $\gamma' = \gamma_{m,0}$ for some $m \in \{1, \dots, n\}$. In particular, γ' is an inner word of I , so

$$\gamma' \in (F \cup R(\beta))^* \setminus (F \cup R(\beta)).$$

Furthermore, $\beta_s \sqsubseteq \beta$, thus $R(\beta) \subseteq R(\beta_s)$, hence $\gamma' \in (F \cup R(\beta_s))^*$. Suppose that $\beta_s \sqsubseteq \gamma'$. Since $\beta_r \sqsubset_{\text{ext}} \beta_s$, there exists $\delta \sqsubset_{\text{ext}} \gamma'$ such that $\beta_r \sqsubseteq_{\text{dil}} \delta$, which contradicts the minimality of γ' . Hence we have proved that

$$\gamma' \in (F \cup R(\beta_s))^* \setminus (F \cup R(\beta_s)).$$

Observe that $e(\gamma') = e(\beta_r) = e(\beta_s)$. $\square$

From now on, until the end of the proof of Theorem 2.9, we fix β , γ , and F that satisfy the conclusion of Lemma 2.35, and we put $u = e(\beta) = e(\gamma)$. We observe that $\gamma \notin R(\beta)$. For a subset G of $R(A)$, we shall say that an inference I from $G \cup R(\beta)$ is *normal*, if $\beta \sqsubseteq \delta$ implies that $\beta \sqsubseteq_{\text{dil}} \delta$, for any $\delta \in \text{rng } I$. We observe that the concept of a normal inference is defined relatively to the (frozen) element β but also to the subset G .

For U , $G \subseteq R(A)$, we define a subset $E(U, G)$ of A and an element $E^+(U, G)$ of L by

$$E(U, G) = \{e(\alpha) : \alpha \in U \cap (G \setminus R(\beta))\},$$

$$E^+(U, G) = \bigvee E(U, G).$$

The main lemma towards the proof of Theorem 2.9 is then the following:

Lemma 2.36. *Let $G \subseteq R(A)$, let I be a normal inference of γ from $G \cup R(\beta)$. If $s \in \omega$ and $\lambda, \lambda_1, \dots, \lambda_s$ are words of I with $\lambda \notin \bigcup_{1 \leq i \leq s} \text{rng } I(\lambda_i)$, then the following inequality holds in L :*

$$e(\lambda) \leq e(\lambda_1) + \dots + e(\lambda_s) + E^+\left(\text{rng } I \setminus \bigcup_{1 \leq i \leq s} \text{rng } I(\lambda_i), G\right). \quad (2.7)$$

Proof. We begin with the following claim.

Claim 1. *Let $G \subseteq R(A)$, let I be a normal inference of γ from $G \cup R(\beta)$. Then the inequality*

$$e(\lambda) \leq u + E^+(\text{rng } I, G) \quad (2.8)$$

holds for any $\lambda \in \text{rng } I$.

Proof. If λ is a free word of I , then either $\lambda \in G \setminus R(\beta)$ in which case $e(\lambda) \leq E^+(\text{rng } I, G)$, or $\lambda \in R(\beta)$, in which case, by the normality assumption, $\beta \sqsubseteq_{\text{dil}} \lambda$, so $e(\lambda) = u$. The general case follows by a straightforward induction on $|I(\lambda)|$. $\square$

The goal of our next claim is to remove u from inequality (2.8).

Claim 2. *Let $G \subseteq R(A)$, let I be a normal inference of γ from $G \cup R(\beta)$. Then the inequality*

$$u \leq E^+(\text{rng } I, G) \quad (2.9)$$

holds.

Proof. If γ is a free word of I , that is, $I = \langle \gamma \rangle$, then $\gamma \in E(\text{rng } I, G)$, so the statement is evident. Now suppose that γ is an inner word of I . Hence, there are nonnegative integers $k \geq 1$ and $s \leq k$ with elements $\gamma_0, \dots, \gamma_k$ of $\text{rng } I$ such that $\gamma_0, \dots, \gamma_{s-1}$ are inner words, $\gamma_s, \dots, \gamma_k$ are free words, and

$$I \vdash \gamma = f_k^*(\gamma_0, \dots, \gamma_k).$$

Since $u = e(\gamma) \neq e(\gamma_i)$ for all $i \in \{0, \dots, k\}$, we get that $\gamma_s, \dots, \gamma_k$ belong to $G \setminus R(\beta)$. Hence, the following inequality holds:

$$e(\gamma) \leq e(\gamma_0) + \dots + e(\gamma_{s-1}) + E^+(\text{rng } I, G). \quad (2.10)$$

In view of Claim 1, (2.10), and the join-semidistributive law, we get

$$\begin{aligned} u + E^+(\text{rng } I, G) &= e(\gamma_0) + \dots + e(\gamma_{s-1}) + E^+(\text{rng } I, G) \\ &= (u \wedge e(\gamma_0)) + \dots + (u \wedge e(\gamma_{s-1})) + E^+(\text{rng } I, G). \end{aligned}$$

Since u is an atom, we get that $u \leq E^+(\text{rng } I, G)$. $\square$

Now we can conclude the proof of Lemma 2.36. By Corollary 2.26, we can assume without loss of generality that $I(\lambda_1), \dots, I(\lambda_s)$ are pairwise disjoint. Put $I' = I[\lambda_1, \dots, \lambda_s]$ and $G' = G \cup \{\lambda_1, \dots, \lambda_s\}$. By Lemma 2.29, I' is an (obviously normal)

inference of γ from G' . Then $\lambda \in \text{rng } I'$, thus, by Claims 1 and 2,

$$e(\lambda) \leq E^+(\text{rng } I', G') \leq e(\lambda_1) + \dots + e(\lambda_s) \\ + E^+\left(\text{rng } I \setminus \bigcup_{1 \leq i \leq s} \text{rng } I(\lambda_i), G\right),$$

which concludes the proof. $\square$

Finally, we shall get a contradiction that proves that the assumption made at the beginning of the proof of Theorem 2.9 is wrong.

This assumption has lead us, see Lemma 2.35, to the existence of $F \in T(L, A)$ and $\beta, \gamma \in R(A)$ such that $e(\beta) = e(\gamma)$ and $\gamma \in (F \cup R(\beta))^* \setminus (F \cup R(\beta))$. By Lemma 2.21, there exists an inference I of γ from $F \cup R(\beta)$. By Lemma 2.33, every word of $R(\beta) \cap \text{rng } I$ dilutes β , that is, I is a normal inference from $F \cup R(\beta)$.

We denote by U the set of inner words of I , and we define finite subsets W_n , for $n \in \omega$, by induction on n , as follows:

$$W_0 = \{\gamma\}.$$

$$W_{n+1} = W_n \cup \{\kappa \frown \langle e(\alpha) \rangle : \kappa \in W_n \text{ and } \alpha \in U \\ \text{such that } \bar{\alpha} \sqsubseteq \kappa \text{ and } \alpha \not\sqsubseteq \kappa\}$$

for all $n \in \omega$. Then we put $W = \bigcup_{n \in \omega} W_n$. We observe that all words of W extend γ .

Lemma 2.37. (i) For any $\kappa \in W$, there exists $\alpha \in U$ such that $e(\kappa) = e(\alpha)$.

(ii) W is finite.

(iii) The relation $\alpha \sqsubseteq \kappa$ holds, for any $\alpha \in U$ and any $\sqsubseteq_{\text{ext}}$ -maximal element κ of W .

Proof. Statement (i) is obvious by the definition of W .

(ii) We claim that there is no $\xi \in R(A)$ such that $|\xi| > |U|$ and $\gamma \frown \xi \in W$. Suppose otherwise, put $l = |\xi|$ and put $\kappa_n = \gamma \frown (\xi|_n)$ for all $n \in \{0, \dots, l\}$, where $\xi|_n$ denotes the restriction of ξ to its first n entries. Observe that $\kappa_n \in W$ for all $n \in \{0, \dots, l\}$. Furthermore, if $n < l$, then there exists $\alpha_n \in U$ such that $\bar{\alpha}_n \sqsubseteq \kappa_n$, $\alpha_n \not\sqsubseteq \kappa_n$, and $\kappa_{n+1} = \kappa_n \frown \langle e(\alpha_n) \rangle$. Since $l > |U|$, there are m, n in $\{0, \dots, l-1\}$ such that $m < n$ and $\alpha_m = \alpha_n$. From $\bar{\alpha}_m \sqsubseteq \kappa_m$ follows that $\alpha_n = \alpha_m \sqsubseteq \kappa_m \frown \langle e(\alpha_m) \rangle = \kappa_{m+1} \sqsubseteq_{\text{ext}} \kappa_n$, which contradicts the fact that $\alpha_n \not\sqsubseteq \kappa_n$. This proves our claim. Therefore, $|W| \leq p^{|U|}$, so W is finite.

(iii) Since both α and κ extend γ , there exists a largest, with respect to $\sqsubseteq_{\text{ext}}$, word v such that $v \sqsubseteq_{\text{ext}} \alpha$ and $v \sqsubseteq \kappa$. Hence, it suffices to prove that $v = \alpha$. Suppose otherwise, and let δ be the successor of v in the $\sqsubseteq_{\text{ext}}$ -interval from v to α , that is, $v \sqsubseteq_{\text{ext}} \delta \sqsubseteq_{\text{ext}} \alpha$ and $\bar{\delta} = v$. It follows from Corollary 2.32 that $\delta \in U$. Moreover, $v = \bar{\delta} \sqsubseteq \kappa$ while $\delta \not\sqsubseteq \kappa$ by the maximality of v . Hence $\kappa \frown \langle e(\delta) \rangle$ belongs to W , which contradicts the maximality of κ . $\square$

We shall get a contradiction proving that all words from W belong to the filter F . Indeed, by assumption, $\gamma \notin F$.

- (1) *Every maximal element of W belongs to F .* Let κ be a maximal element of W . For any $\alpha \in F \cap \text{rng } I$, since there exists, by Lemma 2.31, an I -inference chain from α to γ and since $\alpha \neq \gamma$, $\bar{\alpha}$ is an inner word of I , thus, by Lemma 2.37(iii), $\bar{\alpha} \sqsubseteq \kappa$. If $e(\kappa) = e(\alpha)$ then $\alpha \sqsubseteq \kappa$, whence $\kappa \in F$.

So suppose that $e(\kappa) \neq e(\alpha)$, for all $\alpha \in F \cap \text{rng } I$. In particular, $\kappa \frown \langle e(\alpha) \rangle \in R(A)$. Observe that $F \cap \text{rng } I$ is nonempty, and write $F \cap \text{rng } I = \{\alpha_0, \dots, \alpha_s\}$. By Lemma 2.37(i), there exists $\lambda \in U$ such that $e(\kappa) = e(\lambda)$. Therefore, by Lemma 2.36

$$e(\kappa) = e(\lambda) \leq E^+(\text{rng } I, F) \leq e(\alpha_0) + \dots + e(\alpha_s),$$

whence $s \geq 1$ (because $e(\kappa) \neq e(\alpha_i)$ for all i) and

$$\kappa = f_s(\kappa \frown \langle e(\alpha_0) \rangle, \dots, \kappa \frown \langle e(\alpha_s) \rangle). \quad (2.11)$$

For all $i \in \{0, \dots, k\}$, $\bar{\alpha}_i \sqsubseteq \kappa$, thus $\alpha_i \sqsubseteq \kappa \frown \langle e(\alpha_i) \rangle$; whence $\kappa \frown \langle e(\alpha_i) \rangle \in F$. So it follows from (2.11) that $\kappa \in F$.

- (2) *Let $\kappa \neq \gamma$ in W such that $\kappa \sqsubset \kappa'$ implies that $\kappa' \in F$, for any $\kappa' \in W$. We prove that $\kappa \in F$.* By the definition of W ,

$$\kappa = \bar{\kappa} \frown \langle e(\alpha) \rangle \text{ for some } \alpha \in U \text{ such that}$$

$$\bar{\alpha} \sqsubseteq \bar{\kappa} \text{ and } \alpha \not\sqsubseteq \bar{\kappa}. \quad (2.12)$$

In particular, $\alpha \sqsubseteq \kappa$. Since κ is not maximal in W , there exists an inner word δ of I such that $\delta \not\sqsubseteq \kappa$ and $\bar{\delta} \sqsubseteq \kappa$. Let $\delta_0, \dots, \delta_t$ list all such inner words. If $\lambda \in U$ such that $\lambda \not\sqsubseteq \kappa$, then $\lambda \neq \gamma$, thus $\bar{\lambda}$ is an inner word of I ; if $\bar{\lambda} \sqsubseteq \kappa$, then λ is one of the δ_i , otherwise, we may start again the process at $\bar{\lambda}$ and argue by induction. As a consequence of this,

Every inner word λ of I such that

$$\lambda \not\sqsubseteq \kappa \text{ extends one of the } \delta_i. \quad (2.13)$$

For all $i \in \{0, \dots, t\}$, $\kappa \frown \langle e(\delta_i) \rangle \in W$ (because $\bar{\delta}_i \sqsubseteq \kappa$ and $\delta_i \not\sqsubseteq \kappa$), thus, by the induction hypothesis, $\kappa \frown \langle e(\delta_i) \rangle \in F$. As $\delta_i \not\sqsubseteq \alpha$ (otherwise, $\delta_i \sqsubseteq \kappa$, a contradiction), α does not belong to $\text{rng } I(\delta_0) \cup \dots \cup \text{rng } I(\delta_t)$ (we use Lemma 2.22(i)). Hence, by Lemma 2.36, if we put $X = \text{rng } I \setminus \bigcup_{0 \leq i \leq t} \text{rng } I(\delta_i)$,

$$e(\kappa) = e(\alpha) \leq e(\delta_0) + \dots + e(\delta_t) + E^+(X, F). \quad (2.14)$$

Let $\lambda \in F \cap X$. Then $\bar{\lambda}$ is an inner word of I , and, by Lemma 2.27, $\lambda \in \text{rng } I(\bar{\lambda})$. Suppose that $\delta_i \sqsubseteq_{\text{ext}} \bar{\lambda}$, with $i \in \{0, \dots, t\}$. Then, by Lemma 2.27, $\text{rng } I(\bar{\lambda}) \subseteq \text{rng } I(\delta_i)$, whence $\lambda \in \text{rng } I(\delta_i)$, a contradiction. It follows then from

(2.13) that $\tilde{\lambda} \sqsubseteq \kappa$. Hence, if $e(\lambda) = e(\kappa)$ for some $\lambda \in F \cap X$, then $\lambda \sqsubseteq \kappa$, so $\kappa \in F$. So suppose that $e(\lambda) \neq e(\kappa)$ for all $\lambda \in F \cap X$. Write $X = \{\lambda_1, \dots, \lambda_u\}$. It follows from (2.14) that

$$\begin{aligned} \kappa &= f_{t+u}(\kappa \frown \langle e(\delta_0) \rangle, \dots, \kappa \frown \langle e(\delta_t) \rangle, \\ &\quad \kappa \frown \langle e(\lambda_1) \rangle, \dots, \kappa \frown \langle e(\lambda_u) \rangle), \end{aligned}$$

whence $\kappa \in F$.

By (1) and (2) above, $W \setminus \{\gamma\}$ is contained in F .

(3) Finally, γ is an inner word of I , so

$$\gamma = f_{m-1}(\gamma \frown \langle a_1 \rangle, \dots, \gamma \frown \langle a_t \rangle, \gamma \frown \langle b_1 \rangle, \dots, \gamma \frown \langle b_{m-t} \rangle) \quad (2.15)$$

for some $m \geq 2$ and inner words $\gamma \frown \langle a_1 \rangle, \dots, \gamma \frown \langle a_t \rangle$ and free words $\gamma \frown \langle b_1 \rangle, \dots, \gamma \frown \langle b_{m-t} \rangle \in F$. As $\gamma \frown \langle a_i \rangle$ belongs to W , it belongs to F by item (2) above. Therefore, it follows from (2.15) that $\gamma \in F$.

This contradiction concludes the proof of Theorem 2.9. $\square$

3. Atomistic and strong (nonnecessarily finite) convex geometries

3.1. Atomistic convex geometries

As it was already noted in the introduction, finite convex geometries, from the lattice point of view, are dual to *antimatroids*. In particular, this name indicates that among combinatorial objects, they are antithetical to the well-studied objects called *matroids*. The latter are an abstraction of the idea of linear dependence of vectors in vector spaces. Like finite convex geometries, they can be defined in terms of a closure operator acting on a finite underlying set (see for example [67]). Specifically, a matroid is a finite zero-closure space (X, Φ) satisfying the *exchange property*:

$$x \in \Phi(A \cup \{y\}) \text{ and } x \notin A \text{ imply that } y \in \Phi(A \cup \{x\}),$$

$$\text{for all } x \neq y \text{ in } X \text{ and all closed } A \subseteq X. \quad (\text{EP})$$

A *combinatorial geometry* (see [37]) is defined as an algebraic, atomistic closure space (X, Φ) (see Section 1.2) with Φ satisfying the exchange property. The closure lattice of such a space is a *geometric lattice*, that is, it is algebraic, atomistic, and semimodular. Many examples of geometric lattices appear in connection with various geometrical objects. The subspace lattices of vector spaces as well as the lattices of linear subspaces of projective geometries give the examples of modular geometric lattices. They have been playing a significant role in the development of theory of modular lattices. On the other hand, there exist nonmodular geometric

lattices such as the lattice of affine subspaces of a vector space or the lattice of equivalence relations on a set. The latter were used for special representation of modular and Arguesian lattices [45] (we shall discuss this in more detail in Section 3.8).

Not so much was done for extending the notion of a finite convex geometry to the infinite case. In [10] the lattices satisfying the anti-exchange property (or AEP-lattices), see Section 1.2, were introduced. Like combinatorial geometries, they can also be defined in terms of closure spaces, see Definition 1.6.

In our study of infinite convex geometries, we will be focused mostly on *atomistic* convex geometries. We have already seen in Theorem 1.11 that any finite convex geometry can be “extended” to a finite *atomistic* convex geometry that is defined on the same underlying set. Furthermore, all the interesting classes of closure spaces that we shall mention in the rest of Section 3 are atomistic. Finally, this choice will keep us also closer to the classical notion of a combinatorial geometry.

Theorem 1.9 has the following analogue for the infinite, atomistic case:

Proposition 3.1. *A lattice L is the closure lattice $\text{Cl}(X, \Phi)$ of some atomistic convex geometry (X, Φ) iff L is complete, atomistic, and satisfies $\text{SD}_\vee(\text{At } L)$ (see Definition 1.1), where $\text{At } L$ denotes the set of all atoms of L . In particular, $\text{Cl}(X, \Phi)$ is lower semimodular.*

Proof. The first assertion is evident. To prove the second one, observe that if $a \succ b$ in L , then, by the anti-exchange property, the sets of atoms under a and b differ by a single atom. Then for arbitrary $c \in L$ the corresponding sets of atoms under $c \wedge a$ and $c \wedge b$ differ at most by a single atom. This implies lower semimodularity. $\square$

Observe that there are nonjoin-semidistributive atomistic convex geometries. On the other hand, for a given atomistic convex geometry G , let $S(G)$ denote the join-subsemilattice of G generated by the atoms of G . If $S(G)$ is a sublattice of G , then, by Corollary 1.3, $S(G)$ is join-semidistributive.

The following assertion shows that in contrast with infinite combinatorial geometries that may be of finite height, atomistic convex geometries are either finite or “infinite dimensional”.

Proposition 3.2. (i) *Any join-semidistributive lattice without infinite chains is finite.*
(ii) *Any atomistic convex geometry without infinite chains is finite.*

Note. Claim (i) was proved in [47], see also [27, Theorem 5.59]. Claim (ii) generalizes Theorem 8 from [9] since the requirement of biatomicity is superfluous there. Its proof follows from (i) and Lemma 1.2.

Now we turn to examples to show the diversity of atomistic convex geometries of different nature. Like combinatorial geometries, they emerge in different branches of mathematics.

3.2. Euclidean spaces

Let $\mathbb{E}^n$ denote the n -dimensional Euclidean space over the field of real numbers. A subset $X \subseteq \mathbb{E}^n$ is called *convex* if X contains the segment $[u, v]$ whenever the points u, v belong to X . The *convex hull* of a subset $Y \subseteq \mathbb{E}^n$ is the smallest convex subset of $\mathbb{E}^n$ containing Y .

Proposition 3.3. *The closure space $(\mathbb{E}^n, \text{Co})$, where Co denotes the closure operator of taking the convex hull, is an atomistic convex geometry.*

The closure lattice $\mathbf{Co}(\mathbb{E}^n)$ of this space has been intensively studied in many papers by Bennett (see, for example, [7,8]). This lattice has a large modular core, generated by the affine subspaces of $\mathbb{E}^n$, see [59]. At the same time it contains a large join-semidistributive sublattice, as indicated by the following Theorem 3.4.

Recall that a *convex body* is a convex compact subset of $\mathbb{E}^n$. Evidently, all convex bodies with $\mathbb{E}^n$ added as a top element form a complete lattice $\mathbf{CB}(\mathbb{E}^n)$ that is a sublattice as well as a complete meet-subsemilattice in $\mathbf{Co}(\mathbb{E}^n)$.

Theorem 3.4. *The lattice $\mathbf{CB}(\mathbb{E}^n)$ is join-semidistributive.*

As a matter of fact, $\mathbf{CB}(\mathbb{E}^n)$ is the closure lattice of the atomistic convex geometry $(\mathbb{E}^n, \text{CB})$, where $\text{CB}(Y)$ denotes the closure of the convex hull of $Y \subseteq \mathbb{E}^n$ if Y is bounded, and $\mathbb{E}^n$ otherwise.

The problem of the description of the lattices $\mathbf{CB}(\mathbb{E}^n)$ and their sublattices is posed in [38].

Note. For the lattice $\mathbf{K}(\mathbb{E}^n)$ of polytopes, that is, finitely generated convex sets, of $\mathbb{E}^n$, that is a sublattice of $\mathbf{CB}(\mathbb{E}^n)$, an analogous result was proved in Birkhoff and Bennett [10].

Proof. Any convex body is the closure of the convex hull of the set of its extreme points (see, for example, [22]). Recall that $p \in X$ is an extreme point of a convex set X if $p \notin [p_1, p_2]$ for any $p_1, p_2 \in X \setminus \{p\}$.

Suppose that $V = X \vee Y = X \vee Z$ in $\mathbf{CB}(\mathbb{E}^n)$. As $X \vee Y = \bigcup \{[x, y] : x \in X, y \in Y\}$, any extreme point of V lies either in X or in Y . At the same time it should be in X or in Z , hence it lies either in X or in $Y \cap Z$. This implies that $V = X \vee (Y \cap Z)$. $\square$

On the other hand, $\mathbf{CB}(\mathbb{E}^n)$ fails some of the basic properties of algebraic lattices recorded in Lemma 2.1:

Proposition 3.5. *For any positive integer n , $\mathbf{CB}(\mathbb{E}^n)$ is neither weakly atomic nor dually spatial.*

Proof. Let $\emptyset \neq C \subset D$ be convex bodies and let $d \in D \setminus C$. Consider the segment $[c, d]$ with $c \in C$. As C is closed, one can find $d_1 \in [c, d]$, $d_1 \neq d$, with $d_1 \in D \setminus C$. Then the convex hull F of $C \cup \{d_1\}$ is, since $d \notin F$, a convex body such that $C \subset F \subset D$.

As for any convex body C in $\mathbb{E}^n$ and $x \notin C$ one can find a larger convex body C' with $x \notin C'$, any convex body is an intersection of larger convex bodies. Thus, $\mathbf{CB}(\mathbb{E}^n)$ has no completely meet-irreducible elements. $\square$

3.3. Posets and graphs

For a quasiordered set $(P, \leq)$ (i.e., $\leq$ is a reflexive and transitive binary relation on P), a subset X of P is (order-) *convex*, if $[x, y] = \{t \in P : x \leq t \leq y\}$ is contained in X whenever $x, y \in X$ such that $x \leq y$. Then the map Co that with every subset X of P associates its *convex hull*, that is, the least convex subset of P that contains X , is an algebraic zero-closure operator on P . We denote by $\mathbf{Co}(P) = \mathbf{Co}(P, \leq)$ its closure space. If $\bar{P}$ denotes the quotient of P by the equivalence relation $\equiv$ defined by $x \equiv y$ iff $x \leq y \leq x$, then $\mathbf{Co}(P)$ and $\mathbf{Co}(\bar{P})$ are naturally isomorphic (see [10, Theorem 6]). Furthermore, in that case, the closure map Co is atomistic. The following result, proved in [10, Theorems 6 and 14], implies immediately that if $\leq$ is antisymmetric, then (P, Co) is an atomistic convex geometry:

Theorem 3.6. *The lattice $\mathbf{Co}(P, \leq)$ is join-semidistributive.*

Theorem 3.6 makes it possible to construct join-semidistributive atomistic convex geometries from other types of combinatorial objects. A *directed graph* is, by definition, a pair (V, E) where V is a (not necessarily finite) set (of ‘vertices’) and $E \subseteq V \times V$ is a set (of ‘edges’).

A *path* from $a \in V$ to $b \in V$ is a finite sequence $(v_i)_{0 \leq i \leq n}$ of vertices with $(v_i, v_{i+1}) \in E$ for $0 \leq i < n$. We shall call a subgraph $X \subseteq V$ *convex* if X contains all paths between any two of its points. Then the map Cg that with every subset of V associates the least convex subgraph of P that contains X is an algebraic zero-closure operator on V . We denote by $\mathbf{Cg}(V, E)$ the associated closure lattice.

Let $\leq_E$ be the reflexive and transitive closure of E . Then it is obvious that $\mathbf{Cg}(V, E) = \mathbf{Co}(V, \leq_E)$. Therefore, by using Theorem 3.6, we obtain immediately the following result:

Theorem 3.7. *The lattice $\mathbf{Cg}(V, E)$ of all convex subsets of a graph (V, E) is join-semidistributive.*

Note. The easy consequence of Theorem 3.7 that $\mathbf{Cg}(V)$ is lower semimodular was proved for finite V in [58].

3.4. Lattices of suborders

Consider a pair $(P, \mathcal{L})$ where P is a set and $\mathcal{L} \subseteq P \times P$ is a strict partial order on P . Let $\mathcal{O}$ be the operator of transitive closure defined on $\mathbf{B}(\mathcal{L})$. Then the

closed elements of this closure operator are exactly the partial orders on P contained in $\mathcal{L}$.

Theorem 3.8. *The operator $\mathcal{O}$ satisfies the anti-exchange property, so $(\mathcal{L}, \mathcal{O})$ is an algebraic, atomistic convex geometry. Its closure lattice $\mathcal{O}(\mathcal{L})$ is join-semidistributive iff $(P, \mathcal{L})$ does not contain any infinite bounded chain.*

The last statement of Theorem 3.8 is proved in [62]. As a consequence of Theorem 3.8, we get the result, proved in [14], that $\mathcal{O}(\mathcal{L})$ is lower semimodular.

Proposition 3.9. *If a poset P contains an infinite bounded chain, then $\mathcal{O}(P)$ is not lower continuous.*

Proof. Suppose, for example, that P contains a strictly increasing chain of the form $\langle a_n : 0 \leq n \leq \omega \rangle$. We put $\mathcal{A} = \{(a_i, a_j) : i < j < \omega\}$ and $\mathcal{B}_n = \{(a_i, a_j) : n \leq i < j \leq \omega\}$, for all $n \in \omega$. Then $(a_0, a_\omega) \in \mathcal{A} \vee \mathcal{B}_n$ for all $n \in \omega$, while $(a_0, a_\omega) \notin \mathcal{A} = \mathcal{A} \vee \bigcap_{n \in \omega} \mathcal{B}_n$. $\square$

3.5. Semilattices

Let $(L, \wedge)$ be a meet-semilattice. One can consider three different anti-exchange closures on L . For $X \subseteq L$, we put

$\text{Sub}_\wedge(X)$ = subsemilattice of L generated by X ,

$$\text{Sub}_f(X) = \begin{cases} \text{subsemilattice of } L \text{ generated by } X & \text{if } X \text{ is finite,} \\ L & \text{otherwise,} \end{cases}$$

$\text{Sub}_c(X)$ = complete subsemilattice of L generated by X

(in case L is complete).

In the above definitions, it is convenient to include $\emptyset$ among the (complete) subsemilattices of L .

Theorem 3.10. *Let L be a meet-semilattice (resp. a complete meet-semilattice). Then the closure spaces $(L, \text{Sub}_\wedge)$ and (L, Sub_f) (resp., (L, Sub_c)) are atomistic convex geometries.*

We shall denote by $\mathbf{Sub}_\wedge(L)$, $\mathbf{Sub}_f(L)$, and $\mathbf{Sub}_c(L)$, respectively, the corresponding closure spaces. The necessary and sufficient conditions for these lattices to be join-semidistributive were found in [1]. In particular,

Theorem 3.11. (1) For arbitrary $(L, \wedge)$, the lattice $\mathbf{Sub}_f(L)$ is join-semidistributive.

(2) If L does not contain infinite increasing chains, then both $\mathbf{Sub}_\wedge(L)$ and $\mathbf{Sub}_c(L)$ are join-semidistributive.

We also observe the following analogue of Proposition 3.5:

Proposition 3.12. Let L be a meet-semilattice (resp. a complete meet-semilattice). Then $\mathbf{Sub}_f(L)$ (resp., $\mathbf{Sub}_c(L)$) is weakly atomic but not necessarily dually spatial.

Proof. Let $S \subset T$ be finite (complete) subsemilattices of L and let $t \in T \setminus S$. If T_1 is the finite (complete) subsemilattice of L generated by $S \cup \{t\}$, then $S_1 = T_1 \setminus \{t\}$ is a finite (complete) subsemilattice of L . Hence, $S \subseteq S_1 \prec T_1 \subseteq T$.

Denote by $\mathbb{R}^+$ the complete meet-semilattice of positive real numbers $\mathbb{R}^+$ with natural ordering. Any $X \in \mathbf{Sub}_c(\mathbb{R}^+)$, $X \neq \mathbb{R}^+$, is disjoint from some interval $[r_1, r_2]$ with $r_1 < r_2$. Pick elements r'_1, r'_2 such that $r_1 < r'_1 < r'_2 < r_2$. Then the equality $X = (X \cup \{r'_1\}) \cap (X \cup \{r'_2\})$ shows that X is meet-reducible in $\mathbf{Sub}_\wedge(L)$. The same holds for $X \in \mathbf{Sub}_f(\mathbb{R}^+)$. $\square$

As a consequence, neither $\mathbf{Sub}_f(L)$ nor $\mathbf{Sub}_c(L)$ is necessarily algebraic (see Lemma 2.1). The following examples show that the situation with dual algebraicity is not better.

Example 3.13. Let L be an infinite chain. Then any subset of L is a subsemilattice. It follows that L is the only dually compact element of $\mathbf{Sub}_f(L)$. In particular, $\mathbf{Sub}_f(L)$ is not dually algebraic, while it is, of course, lower continuous.

The next example is taken from the description of dually algebraic lattices $\mathbf{Sub}_\wedge(L)$ in [1].

Example 3.14. Let $L = (\omega + 1) \cup \{t\}$, where the ordinal $\omega + 1$ is endowed with its natural ordering and t is a new element such that $0 < t < \omega$ and t is incomparable with all positive integers. So L is a complete meet-semilattice. Then neither $\mathbf{Sub}_\wedge(L)$ nor $\mathbf{Sub}_c(L)$ is dually algebraic (it is not even lower continuous).

3.6. Universal Horn logic and lattices of algebraic subsets

We recall that $\mathcal{L}_q(\mathcal{K})$ denotes the lattice of all subquasivarieties of a given quasivariety $\mathcal{K}$. If $\mathcal{L}_q(\mathcal{K})$ is atomistic then it is an atomistic convex geometry since any quasivariety lattice is join-semidistributive. But the study of such lattices has brought a wider class of atomistic convex geometries. We have already mentioned the fact, proved in [35], that for any algebraic lattice A , the lattice $\mathbf{Sp}(A)$ is isomorphic to $\mathcal{L}_q(\mathcal{K})$ for some quasivariety $\mathcal{K}$ of relational structures.

For arbitrary complete lattices, we can state the following result that was kindly offered to us by F. Wehrung:

Proposition 3.15. *Let L be a complete lattice. Then $S_p(L)$ satisfies $SD_{\vee}(At L)$ (see Definition 1.1). In particular, $S_p(L)$ is the closure lattice of an atomistic convex geometry.*

Proof. We put $L^{\perp} = L \setminus \{1\}$, and, for a subset X of $L^{\perp}$, we define $S_p(X) = \bar{X} \setminus \{1\}$, where $\bar{X}$ denotes the algebraic subset of L generated by X . Then S_p is an atomistic closure operator on $B(L^{\perp})$, and the rule $X \mapsto X \cup \{1\}$ defines an isomorphism from $Cl(L^{\perp}, S_p)$ onto $S_p(L)$.

To conclude the proof, it suffices to verify that $(L^{\perp}, S_p)$ satisfies (AEP). Let X be a closed subset of $L^{\perp}$. It is convenient to observe first the following preliminary claim. For any $c \in L^{\perp}$, the subset $X \cup [0, c]$ is a closed subset of $L^{\perp}$ and it contains $X \cup \{c\}$, whence $S_p(X \cup \{c\}) \subseteq X \cup [0, c]$. Now let $a \neq b$ be elements of $L^{\perp} \setminus X$. Suppose that $S_p(X \cup \{a\}) = S_p(X \cup \{b\})$. By the claim above, $a \in X \cup [0, b]$ and $b \in X \cup [0, a]$, hence, since $a, b \notin X$, we obtain that $a = b$, a contradiction. $\square$

Remark 3.16. It follows from this and from Lemma 1.2 that the join-semidistributive law

$$X \vee Y = X \vee Z \Rightarrow X \vee Y = X \vee (Y \cap Z)$$

holds in $S_p(L)$ for arbitrary X and finite Y, Z .

Of course, for L complete and upper continuous, a stronger statement holds, namely, $S_p(L)$ is join-semidistributive, see Theorem 2.2(i).

Our next result gives, for a complete upper continuous lattice A , a convenient description of the completely meet-irreducible elements of $S_p(A)$:

Lemma 3.17. *Let A be a complete, upper continuous lattice. We put*

$$[a \rightarrow b] = \{x \in A : a \leq x \Rightarrow b \leq x\}, \quad \text{for all } a, b \in A.$$

Then the completely meet-irreducible elements of $S_p(A)$ are exactly the subsets of the form $[a \rightarrow b]$, where $a < b$ in A and a is compact. Furthermore, the upper cover of $[a \rightarrow b]$ in $S_p(A)$ is $[a \rightarrow b] \cup \{a\}$.

Note. By Theorem 2.2(i), $S_p(A)$ is lower continuous, hence all its completely meet-irreducible elements are dually compact.

Proof. The verification of the fact that $[a \rightarrow b]$ is a completely meet-irreducible element of $S_p(A)$, with upper cover $[a \rightarrow b] \cup \{a\}$, is straightforward. Conversely, let P be a completely meet-irreducible element of $S_p(A)$. We put $a = \bigwedge (A \setminus P)$. For all $x \in A$, we denote by $\bar{x}$ the least element of P above x .

Claim 1. *The element a belongs to $A \setminus P$, and it is compact. Furthermore, $P \cup \{a\}$ is the (unique) upper cover of P in $\mathbf{S}_p(A)$.*

Proof. Suppose that $a \in P$. Since P is completely meet-irreducible, in order to reach a contradiction, it is sufficient to prove that $P = \bigcap_{u \in A \setminus P} P[u]$, where we define $P[u]$ as the algebraic subset of A generated by $P \cup \{u\}$. We observe that

$$P[u] = P \cup \{u \wedge x : x \in P\} = P \cup \{x \in A : x = u \wedge \bar{x}\}.$$

If x is an element of $\bigcap_{u \in A \setminus P} P[u] \setminus P$, then $x = \bar{x} \wedge u$ for all $u \in A \setminus P$, whence $x = \bar{x} \wedge a$. But from $\bar{x} \in P$ and $a \in P$ follows that $x \in P$, a contradiction.

Hence, $a \notin P$, while $x \in P$ for all $x < a$. It follows immediately that $P \cup \{a\}$ belongs to $\mathbf{S}_p(A)$, from which the last statement of the claim follows.

Finally, let I an up-directed poset and let $(a_i)_{i \in I}$ be an increasing family of elements of A such that $a \leq \bigvee_{i \in I} a_i$. Suppose that $a \not\leq a_i$ for all $i \in I$, that is, $a \wedge a_i < a$. Hence $a \wedge a_i \in P$, whence, by using the upper continuity of A , $a = a \wedge \bigvee_{i \in I} a_i \in P$, a contradiction. Therefore, a is compact. $\square$

Claim 2. *The equality $a = \bar{a} \wedge x$ holds, for all $x \in A \setminus P$.*

Proof. From $P \subset P[x]$ and from Claim 1 follows that $a \in P[x] \setminus P$, that is, $a = \bar{a} \wedge x$. $\square$

Claim 3. $P = [a \rightarrow \bar{a}]$.

Proof. The containment $P \subseteq [a \rightarrow \bar{a}]$ follows immediately from the definition of $\bar{a}$. Conversely, let $x \in [a \rightarrow \bar{a}]$, suppose that $x \notin P$. From the definition of a follows that $a \leq x$, thus $\bar{a} \leq x$ by assumption, thus $a = \bar{a}$ by Claim 2, a contradiction. $\square$

Suppose that there exists $x \in A$ such that $a < x < \bar{a}$. From Claim 3 follows that $x \notin P$, then from Claim 2 follows that $a = x$, a contradiction. Hence, $a < \bar{a}$, which concludes the proof. $\square$

As an immediate corollary, even for an algebraic lattice A , the lattice $\mathbf{S}_p(A)$ does not need to be dually spatial, even though, according to Theorem 2.2(ii), it is dually algebraic:

Corollary 3.18. *Let A be the ideal lattice of a dense chain with zero. Then $\mathbf{S}_p(A)$ has no completely meet-irreducible elements.*

Proposition 3.19. *Let X be a set. Then the closure lattice $\mathbf{S}_p(B(X))$ is dually spatial.*

Note. Observe that if A is an arbitrary algebraic lattice and X is the set of its compact elements, then $\mathbf{S}_p(A)$ is a principal ideal in $\mathbf{S}_p(B(X))$, see [35].

Proof. It follows from Lemma 3.17 that the completely meet-irreducible elements of $\mathbf{S}_p(B(X))$ are the $D(C, t) = [C \rightarrow C \cup \{t\}]$, for a finite subset C of X and an element t of $X \setminus C$. For arbitrary $\mathcal{Z} \in \mathbf{S}_p(B(X))$, we denote by Z_C the least element of $\mathcal{Z}$ above C , for every finite subset C of X . Then it is easy to verify that

$$\mathcal{Z} = \bigcap \{D(C, t) : C \subseteq X \text{ finite, } t \in Z_C \setminus C\},$$

whence $\mathcal{Z}$ is a meet of completely meet-irreducible elements of $\mathbf{S}_p(B(X))$. $\square$

It is interesting to study whether a given lattice of the form $\mathbf{S}_p(A)$ can be decomposed as a direct product of directly indecomposable factors. In Wehrung [66], it is proved that for any complete Boolean lattice B , the lattice $\mathbf{S}_p(B)$ is subdirectly irreducible (thus directly indecomposable). On the other hand, if A is defined as the ideal lattice of the real unit interval $[0, 1]$, then $\mathbf{S}_p(A)$ cannot be decomposed as a direct product of directly indecomposable factors, see [5]. Some more information about direct decompositions can be found in Proposition 3.20.

3.7. Defining strong convex geometries

Table 1 gives a list of the principal classes of atomistic convex geometries mentioned in this paper. All the corresponding lattices L are atomistic, complete, and satisfy the weak join-semidistributive law $\text{SD}_\vee(\text{At } L)$.

The list below provides some additional explanation for the entries of Table 1.

- (1) See Proposition 3.5.
- (2) For A the ideal lattice of any dense chain with zero, $\mathbf{S}_p(A)$ is not even dually spatial, see Corollary 3.18.
- (3) See Proposition 3.12.

Table 1
Lattices and closure spaces

Lattice Closure space	$\mathbf{Co}(\mathbb{E}^n)$ ($\mathbb{E}^n, \text{Co}$)	$\mathbf{CB}(\mathbb{E}^n)$ ($\mathbb{E}^n, \text{CB}$)	$\mathbf{Co}(P)$ (P, Co)	$\mathcal{O}(\mathcal{L})$ ($\mathcal{L}, \mathcal{O}$)	$\mathbf{S}_p(A)$ ($A \setminus \{1\}, \mathbf{S}_p$)	$\mathbf{Sub}_\wedge(L)$ ($L, \text{Sub}_\wedge$)	$\mathbf{Sub}_f(L)$ (L, Sub_f)	$\mathbf{Sub}_c(L)$ (L, Sub_c)
Algebraic	Yes	No ⁽¹⁾	Yes	Yes	No ⁽²⁾	Yes	No ⁽³⁾	No ⁽³⁾
Dually algebraic	No ⁽⁴⁾	No ⁽⁵⁾	No ⁽⁴⁾	No ⁽⁶⁾	Yes (A algebraic) No ⁽⁷⁾ (general case)	No ⁽⁸⁾	No ⁽⁹⁾	No ⁽⁸⁾
Weakly atomic	Yes	No ⁽¹⁾	Yes	Yes	Yes	Yes	Yes	Yes
$\text{SD}_\vee$	No ⁽¹⁰⁾	Yes	Yes	No ⁽¹¹⁾	Yes (A upper continuous) No ⁽¹²⁾ (general case)	No ⁽¹³⁾	Yes	No ⁽¹³⁾

- (4) The lattice is not lower continuous. Take $P = \omega$, put $X = \{0\}$ and $Y_n = [n, \infty)$ for all $n \in \omega$. Then $X \vee Y_n = Y_0$ for all n , while $X \vee \bigcap_{n \in \omega} Y_n = X$.
- (5) The top element is the only dually compact one.
- (6) $\mathcal{O}(\mathcal{L})$ is not lower continuous as a rule, see Proposition 3.9.
- (7) See Example 2.3.
- (8) Neither $\mathbf{Sub}_\wedge(L)$ nor $\mathbf{Sub}_c(L)$ need to be lower continuous, see Example 3.14.
- (9) The lattice $\mathbf{Sub}_f(L)$ is lower continuous but it need not be dually algebraic, see Example 3.13.
- (10) For $n \geq 2$, see, for example, [10, p. 234].
- (11) See Theorem 3.8 and [62].
- (12) If A is an upper continuous complete lattice, then $\mathbf{S}_p(A)$ is join-semidistributive, see Theorem 2.2(i). For non upper continuous A , this property may fail, see Theorem 2.8.
- (13) See Theorem 3.11 and [1].

It seems that the notion of atomistic convex geometry is too general to achieve a satisfactory theory. On the other hand, the additional requirement for a closure operator to be algebraic as in combinatorial geometry would imply the loss of some important examples. For instance, the lattice $\mathbf{S}_p(A)$ of algebraic subsets of an algebraic lattice A is not algebraic while it is dually algebraic. We believe that the general definition of a strong convex geometry should be well balanced in the following sense:

- (1) it should include a rich list of examples, in particular among those listed in Table 1;
- (2) it should be restrictive enough to imply reasonably strong structural results.

In particular, the importance of the example of $\mathbf{S}_p(A)$ makes it impossible to define a strong convex geometry to be algebraic, in contrast with the case of combinatorial geometries. On the other hand, we observe that the properties to be dually spatial and weakly atomic are shared by algebraic lattices (see Lemma 2.1) as well as the special types of $\mathbf{S}_p(A)$ considered in Proposition 3.19.

It turns out that complete atomistic dually spatial lattices inherit the decomposition properties of algebraic lattices. For example, the following statement is a consequence of a result in [66], thus extending Libkin's decomposition Theorem [51] to the dually spatial case:

Proposition 3.20. *Any complete atomistic dually spatial lattice can be decomposed as a direct product of directly indecomposable lattices.*

Among the results that one could hope for in (2) would be, for example, results of *spatial theory* of join-semidistributive lattices, that would be, ideally, counterparts of the *modular* case established in [41]. In view of Example 3.25, this is probably not trivial to achieve.

On the other hand, the atomistic weakly atomic convex geometries carry the feature of finite convex geometry expressed in property (4) of Theorem 1.7, which is often considered as the definition of a convex geometry in combinatorics. Basically, weak atomicity says in this context that there should exist *plenty* of closed subsets that can be extended by a single point to another closed subset. We also believe that lower semimodularity should be a sound property in strong convex geometry, so it requires an abundance of covers in the lattice of closed sets.

Thus we propose the following definition of a strong convex geometry:

Definition 3.21. A closure space (X, Φ) is

- (i) *almost algebraic*, if its closure lattice is weakly atomic and dually spatial (see the statement of Lemma 2.1).
- (ii) a *strong convex geometry*, if it is an almost algebraic atomistic convex geometry.

In particular, the following “implications” follow right away from Lemma 2.1:

algebraic atomistic convex geometry

$\Rightarrow$ strong convex geometry

$\Rightarrow$ atomistic convex geometry.

We observe that Table 1 provides numerous examples that show that the implications above cannot be reversed.

3.8. Embedding theorems

In [29] it was proved that any complemented modular lattice can be embedded into a projective geometry. The latter may be viewed as the (modular) lattice of closed sets of the underlying projective space. This cannot be done for arbitrary modular lattices, see, for example, [18] for a counterexample.

Later on, the idea of ordered projective spaces whose subspace lattices are modular and spatial was developed, see [24]. It allowed to prove the following embedding result, see [23].

Theorem 3.22. *Any modular lattice can be embedded into an algebraic spatial modular lattice.*

Our Theorem 2.4 together with observations about the lattices $S_p(A)$ and $S_p(B(X))$ made in the previous subsection can be viewed as an analogous result applied to join-semidistributive lattices and atomistic convex geometries.

Theorem 3.23. *Any finitely presented join-semidistributive lattice can be embedded into a dually algebraic, dually spatial, join-semidistributive, atomistic convex geometry.*

Proof. By Theorem 2.4, we can embed any finitely presented join-semidistributive lattice into $S_p(A)$ for some algebraic lattice A . Furthermore, $S_p(A) \leq S_p(B(X))$ where X is the set of compact elements of A . It is proved in Theorem 2.2 that $S_p(B(X))$ is both join-semidistributive and dually algebraic. By Proposition 3.19, $S_p(B(X))$ is dually spatial. $\square$

The following example shows that Theorem 3.23 cannot be extended from finitely presented to finitely generated join-semidistributive lattices:

Example 3.24. There exists a four-generated join-semidistributive lattice that cannot be embedded into any join-semidistributive lower continuous (in particular, dually algebraic) lattice.

Let us consider the lattice L_4 depicted on the left half of Fig. 2 (see also [44, Fig. 4.6(i)]). We observe that L_4 is join-semidistributive. Let M be a join-semidistributive lower continuous lattice extending L_4 . Put $\bar{b} = \bigwedge_{i \geq 1} b_i$. If $\bar{b} = b$, then $d = d \vee b = d \vee (\bigwedge_{i \geq 1} b_i) = \bigwedge_{i \geq 1} (d \vee b_i) = b_1$, a contradiction.

Thus, we should have $\bar{b} > b$. Then $\bar{b} \not\leq a$. Set $\bar{c} = \bigwedge_{i \geq 1} c_i$. Clearly, $\bar{b} \wedge \bar{c} = 0$. Let us prove that $a \vee \bar{c} = a \vee \bar{b}$. Indeed, $c_i \leq a \vee b_i$ for all $i \geq 1$, hence

$$\bar{c} \leq \bigwedge_{i \geq 1} (a \vee b_i) = a \vee \bigwedge_{i \geq 1} b_i = a \vee \bar{b}.$$

The proof of the converse inequality is similar. Finally, it follows from the join-semidistributivity of M that $a \vee \bar{b} = a \vee (\bar{b} \wedge \bar{c}) = a$, hence $\bar{b} \leq a$, a contradiction.

The following example, communicated to us by F. Wehrung, satisfies an even stronger incompleteness property but it is not finitely generated.

Example 3.25. There exists a join-semidistributive lattice that cannot be embedded into any complete join-semidistributive lattice.

Proof. Let E be the lattice diagrammed on the right half of Fig. 2. Then E is join-semidistributive. Suppose that E embeds into a join-semidistributive lattice L in which every bounded countable sequence admits a supremum. We put

$$a = \bigvee_{n \in \omega} a_n, \quad b = \bigvee_{n \in \omega} b_n.$$

From the fact that $a_n < b_n \vee c$ and $b_n < a_{n+1} \vee c$ for all $n \in \omega$ follows that $a \vee c = b \vee c$, hence, since L is join-semidistributive, $a \leq (a \wedge b) \vee c$. However, $a \leq u$, $b \leq v$, and $u \wedge v = 0$, hence $a_1 < a \leq (u \wedge v) \vee c = c$, a contradiction. $\square$

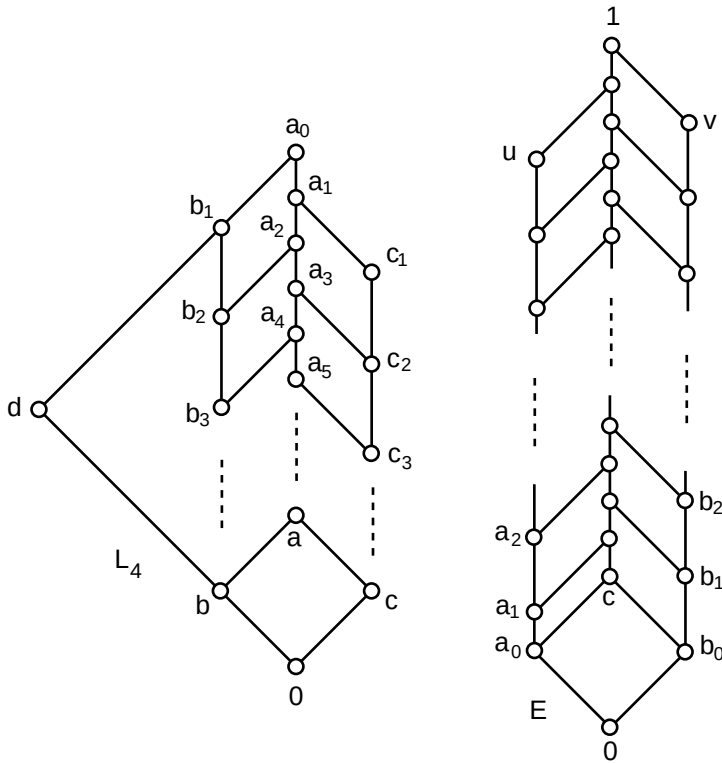


Fig. 2.

What can be said about embedding arbitrary join-semidistributive lattices? Recall that a lattice L is called *biatomic* (see [9]) if it is atomic, that is, any element contains an atom, and for any atom $x \in L$ and $u, v \in L$, if $x \leq u \vee v$ then $x \leq p \vee q$ for some atoms $p \leq u$ and $q \leq v$. This property plays an essential role in the theory of combinatorial geometries, for, as was proved in [9], *the closure lattice of a combinatorial geometry is biatomic iff it is modular*. It seems that biatomicity is much less restrictive for convex geometries. It is amazing that the atomistic convex geometries in *all but one* of the examples given in Sections 3.2–3.6 are biatomic! The only exception is the lattice $\mathcal{O}(\mathcal{L})$ of orders contained in a given order $(X, \mathcal{L})$. So it is not surprising that biatomic atomistic convex geometries give room for arbitrary join semidistributive lattices.

Theorem 3.26. *Any join-semidistributive lattice can be embedded into an atomistic, algebraic, biatomic convex geometry.*

Proof. By Corollary 2.7, the quasivariety of join-semidistributive lattice is generated by the lattices $S_p(A)$ with algebraic A . These lattices are biatomic. This can be either

proved directly or deduced from the fact that they are isomorphic to some lattices of quasivarieties which are always biatomic [4]. Observe that the property of an element of a lattice to be an atom is a first-order property. Hence, the biatomicity is a first order property. Further, let ξ be the following sentence:

$$\xi \Leftrightarrow \forall x, y (x < y \Rightarrow \exists z ((z \text{ is an atom}) \ \& \ (z \leq y) \ \& \ (z \not\leq x))).$$

Then a lattice L with zero satisfies ξ iff L is atomistic. Observe that L may not be complete.

Now let L be an arbitrary join-semidistributive lattice. By Theorems 1.4 and 2.4, $L \in \underline{SP}_u\{\mathbf{S}_p(A) : A \text{ algebraic lattice}\}$. It follows that L can be embedded into an atomistic (generally noncomplete), biatomic, and join-semidistributive lattice L_1 . Let A be the set of atoms of L_1 , and let $\mathbf{Bi}(A)$ be the lattice of all subsets Y of A such that $c \leq a \vee b$ and $a, b \in Y$ implies that $c \in Y$, for all $a, b, c \in A$. Then $\mathbf{Bi}(A)$ is an algebraic subset of the powerset of A , let $X \mapsto \mathbf{Bi}(X)$ be the associated closure operator on A . It is obviously an algebraic closure operator. It follows easily from the biatomicity of L_1 that the join of elements in $\mathbf{Bi}(A)$ has the form:

$$X +_{\mathbf{Bi}} Y = \{z \in A : \exists (x, y) \in X \times Y, z \leq x \vee y\},$$

$$\text{for all } X, Y \in \mathbf{Bi}(A).$$

This implies immediately that $\mathbf{Bi}(A)$ is biatomic.

Let $X \in \mathbf{Bi}(A)$, $a \neq b$ in A , $a \in \mathbf{Bi}(X \cup \{b\}) \setminus X$. Then $a \leq_{L_1} x \vee b$ for some $x \in X$. Assume that $b \in \mathbf{Bi}(X \cup \{a\})$, then $b \leq_{L_1} x' \vee a$ for some $x' \in X$. We get $(x \vee x') \vee a = (x \vee x') \vee b = c$, which implies that $c = x \vee x'$ since L_1 is join-semidistributive, whence $a, b \leq x \vee x'$. Together with $X \in \mathbf{Bi}(A)$, this implies that $a \in X$, a contradiction. Hence, $\mathbf{Bi}$ is an anti-exchange closure operator.

It remains to prove that L_1 embeds into $\mathbf{Bi}(A)$. Define a map ϕ from L_1 to $\mathbf{Bi}(A)$ by the rule $\phi(x) = \{a \in A : a \leq x\}$, for all $x \in L_1$. Clearly, $\phi(x) \in \mathbf{Bi}(A)$ for any $x \in L_1$. As L_1 is atomistic, ϕ is injective. Moreover, ϕ preserves all existing meets. Now let $x, y \in L_1$. Then, by using the biatomicity of L_1 ,

$$\begin{aligned} \phi(x \vee y) &= \{a \in A : a \leq x \vee y\} \\ &= \{a \in A : \exists u, v \in A, u \leq x, v \leq y, \text{ and } a \leq u \vee v\} \\ &= \phi(x) +_{\mathbf{Bi}} \phi(y), \end{aligned}$$

for all $x, y \in L_1$, which concludes the proof. $\square$

The embedding of an arbitrary atomistic biatomic and join-semidistributive lattice into an atomistic, algebraic, biatomic convex geometry given in the previous proof does not preserve the join-semidistributive law in general. For example, by this construction, $\mathbf{S}_p(A)$ is embedded into $\mathbf{Sub}_{\wedge}(A)$ which is generally not join-semidistributive, see [1]. Another example gives the lattice $\mathbf{CB}(\mathbb{E}^n)$ of convex bodies

of $\mathbb{E}^n$ as well as its noncomplete sublattice $\mathbf{K}(\mathbb{E}^n)$ of convex polytopes, because they are both embedded in the same algebraic lattice $\mathbf{Co}(\mathbb{E}^n)$ of convex subsets of $\mathbb{E}^n$ where the join-semidistributive law fails. Because of Example 3.25, one cannot hope to be able to embed an arbitrary join-semidistributive lattice into a complete join-semidistributive lattice, let alone a join-semidistributive convex geometry. Still, the following problem remains open:

Problem 2. Can every finitely generated join-semidistributive lattice be embedded into some complete join-semidistributive lattice?

The join-semidistributive lattice of Example 3.25 cannot be embedded into any complete join-semidistributive lattice, but it is not finitely generated.

Let us see now what is known about lattices which are embeddable into members of given classes of *finite* atomistic convex geometries. By Theorem 1.11, any finite join-semidistributive lattice can be embedded into a finite atomistic convex geometry. This result is the best possible one because finite convex geometries are join-semidistributive as well. However, it would be interesting to find out which join-semidistributive lattices can be embedded into some *specific* finite convex geometries. In particular, *given a class $\mathcal{C}$ of finite join-semidistributive lattices, what are the lattices that can be embedded into some member of $\mathcal{C}$?*

One such subclass is well known. Recall [54] that a finitely generated lattice L is *lower bounded*, if there exists a homomorphism $g: F_L(n) \twoheadrightarrow L$ from a finitely generated free lattice onto L such that any element $x \in L$ has a least pre-image. In [6], the definition was extended, calling a lattice A *lower bounded*, if every finitely generated sublattice of A is lower bounded.

It is known that lower bounded lattices inherit the join-semidistributive law from free lattices and form a proper subclass in $SD_\vee$, see [27] for the most comprehensive account. Not all finite convex geometries are lower bounded. For example the lattice $\mathbf{Co}(P)$ of convex subsets of a finite poset P is not lower bounded unless P does not contain any four-element chain. Some other types of convex geometries are not only lower bounded but contain any finite lower bounded lattice as a sublattice, which can be seen in the next theorem.

Theorem 3.27. *Let L be a finite lattice. Then the following are equivalent:*

- (i) L is lower bounded;
- (ii) L can be embedded into $\mathbf{Sub}_\wedge(A)$ for some finite (semi)lattice A ;
- (iii) L can be embedded into $\mathcal{O}(\mathcal{L})$ for some order $(X, \mathcal{L})$ on a finite set X .

Proof. Implications (i) $\Leftrightarrow$ (ii) follow from results in [3,61], while (i) $\Leftrightarrow$ (iii) follow from results in [63]. $\square$

Observe that it was proved by Dilworth in the 1940s [16] that any finite lattice is embedded into the closure lattice of some finite combinatorial geometry. Four decades later, Pudlák and Tůma [60] strengthened this result by proving that any

finite lattice is isomorphic to a sublattice of the lattice $\mathbf{Eq}(X)$ of equivalence relations on a finite set X . It follows from Theorem 3.27 that finite $\mathbf{Sub}_\wedge(A)$ and $\mathcal{O}(\mathcal{L})$ cannot serve as “universal” convex geometries, that is, those containing all finite join-semidistributive lattices as sublattices.

Not much is known about the sublattices of the lattices $\mathbf{Co}(P)$, for a finite poset P , or $\mathbf{Co}(\mathbb{E}^n, X)$, for a finite set X of points in $\mathbb{E}^n$.

Problem 3. Is there a special type of finite convex geometry that contains all finite join-semidistributive lattices as sublattices? In particular, describe the subclasses of finite join-semidistributive lattices embeddable into finite lattices of the form $\mathbf{Co}(P)$ or $\mathbf{Co}(\mathbb{E}^n, X)$.

Notice that the representation problem was solved for some finite convex geometries of the above forms. The finite lattices representable as $\mathbf{Sub}_\wedge(A)$, for some finite semilattice A , were characterized in [2]. In [10], the description of lattices of the form $\mathbf{Co}(P)$ is given for any poset P .

In this regard it is worthwhile to mention the problem of Jamison about which convex geometries are representable as $\mathbf{Co}(\mathbb{E}^2, X)$ for some X . Some approaches to the solution can be found in [68].

One could also ask the following related question:

Problem 4. Can any finite join-semidistributive lattice be embedded into some finite biatomic atomistic convex geometry?

Theorems 2.8, 2.4, and 3.27 provide now an interesting analogy with Jónsson’s three-type embeddings into lattices $\mathbf{Eq}(X)$ of equivalence relations. More precisely, Jónsson proved [45] that (i) any lattice has type-3 embedding into some $\mathbf{Eq}(X)$, (ii) the lattices with type-2 embedding compose the class of all modular lattices, and (iii) the lattices having type-1 embedding are Arguesian.

Our theorems give a classification of lattices embeddable into $\mathbf{S}_p(A)$ with respect to the properties of complete lattice A : (i) any lattice is embeddable into $\mathbf{S}_p(A)$ with dually algebraic A , (ii) any finitely presented join-semidistributive lattice is a sublattice of $\mathbf{S}_p(A)$ with A being both algebraic and dually algebraic, and (iii) the sublattices of $\mathbf{S}_p(A)$ with finite A compose a class of finite lower bounded lattices. In view of Theorem 2.2, this classification is not complete yet:

Problem 5. Which join-semidistributive lattices can be embedded into $\mathbf{S}_p(A)$ for some complete and upper continuous lattice A ?

As proved in [33], for any upper continuous lattice A , the lattice $\mathbf{S}_p(A)$ is lower continuous. Thus, Example 3.25 shows that the lattices embeddable into $\mathbf{S}_p(A)$ with upper continuous A form a proper subclass of join-semidistributive lattices.

To strengthen the similarity between the two classifications above, one could observe that the class of lower bounded lattices plays in some sense an analogous role in $\mathbf{SD}_\vee$ as Arguesian lattices do in the class $\mathbf{M}$ of modular lattices (see more details in [28]).

Note that as for combinatorial geometries, we can claim that *every* lattice can be embedded into some atomistic convex geometry. Indeed, by Theorem 2.8, every lattice can be embedded into a lattice of the form $\mathbf{S}_p(T)$ for some dually algebraic lattice T , while, by Proposition 3.15, $\mathbf{S}_p(T)$ is the closure lattice of a convex geometry. In fact, every lattice can be embedded into even a *strong* convex geometry, as the following result by Bredikhin and Schein [11] shows:

Theorem 3.28. *Any lattice L can be embedded into the algebraic, atomistic convex geometry $\mathcal{O}(\mathcal{L})$ of all orders containing a given order $(X, \mathcal{L})$, with $|X| = \max\{\aleph_0, |L|\}$.*

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